

Initiating Search

October 16, 2025, 3:42 PM

• Search: Substitution of Aromatic Halides with Nitrogen Nucleophiles

Filtered By:

Yield: 90-100%, 80-89%, 70-79%
Document Type: Journal
Language: English
Publication Year: 2020 to 2022
Publication Name: Journal of Medicinal Chemistry, European Journal of Medicinal Chemistry, Journal of Organic Chemistry, Organic Letters, Angewandte Chemie, International Edition, Journal of the American Chemical Society, Bioorganic & Medicinal Chemistry Letters, ACS Catalysis, Organic Chemistry Frontiers, Green Chemistry, Nature (London, United Kingdom), Nature Communications, Chem, Science (Washington, DC, United States), Nature Catalysis, CCS Chemistry

Search Tasks

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Exported: Returned Reaction Results + Filters (10,074)	 Reactions	View Results

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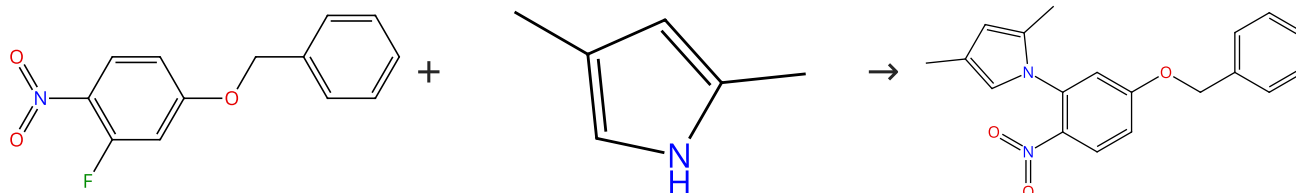
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Reactions (100)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (54)

Suppliers (77)

31-172-CAS-23258076

Steps: 1 Yield: 100%

Enantioselective Construction of Quinoxaline-Based Heterobiphenyls and P,N-Ligands via Chirality Transfer Strategy

By: Gao, Zeng; et al

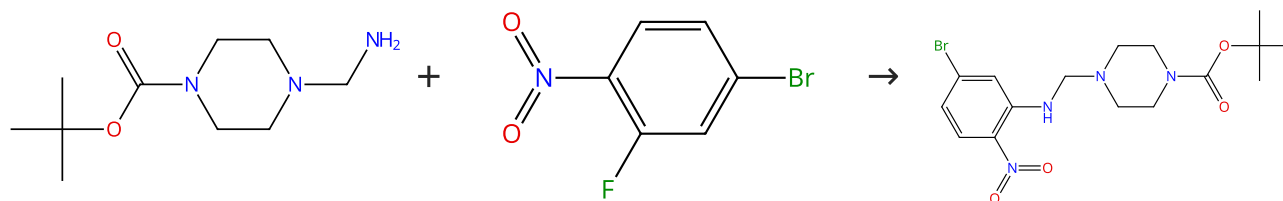
Organic Letters (2021), 23(4), 1181-1187.

1.1 Reagents: Sodium hydroxide
Solvents: Dimethyl sulfoxide; 2 h, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (98)

31-614-CAS-34289474

Steps: 1 Yield: 100%

Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase

By: Kozlova, Arina; et al

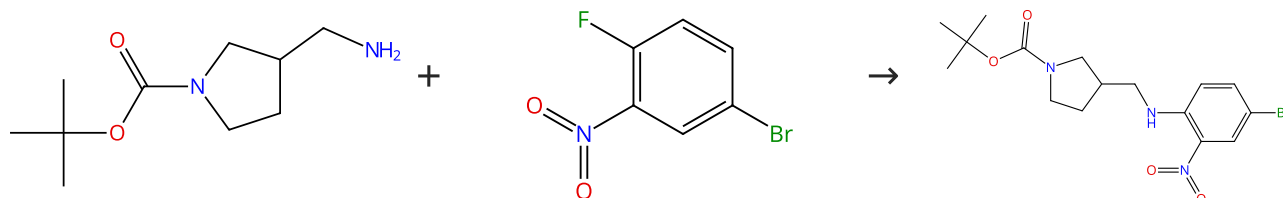
Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

1.1 Reagents: Triethylamine
Solvents: Ethanol; 2.5 h, reflux

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (74)

Suppliers (90)

31-614-CAS-34289437

Steps: 1 Yield: 100%

Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase

By: Kozlova, Arina; et al

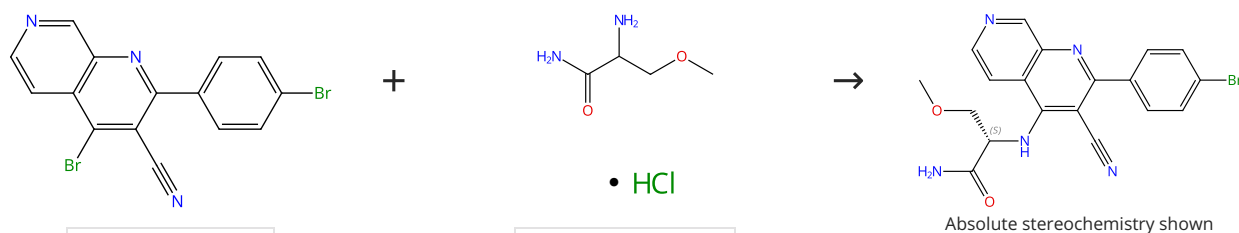
Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

1.1 Reagents: Triethylamine
Solvents: Ethanol; 2.5 h, reflux

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (2)

Suppliers (35)

31-614-CAS-40188236

Steps: 1 Yield: 100%

Discovery and Characterization of the Potent and Highly Selective 1,7-Naphthyridine-Based Inhibitors BAY-091 and BAY-297 of the Kinase PIP4K2A

1.1 **Reagents:** Triethylamine
Solvents: Dimethylformamide; 24 h, rt

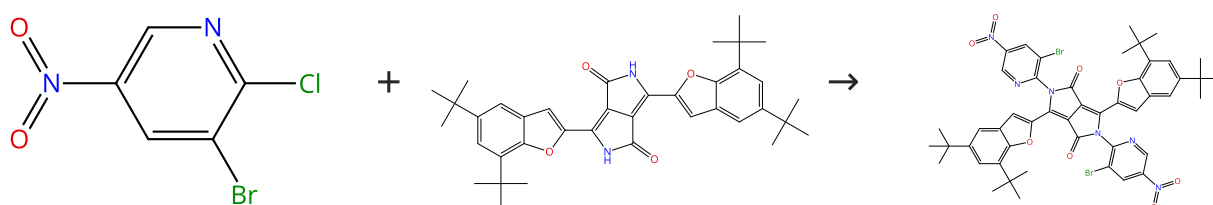
Experimental Protocols

By: Wortmann, Lars; et al

Journal of Medicinal Chemistry (2021), 64(21), 15883-15911.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

31-172-CAS-22502586

Steps: 1 Yield: 100%

How To Make Nitroaromatic Compounds Glow: Next-Generation Large X-Shaped, Centrosymmetric Diketopyrrolopyrroles

1.1 **Reagents:** Potassium fluoride
Solvents: Dimethylformamide; 3 h, 100 °C; 100 °C → 70 °C

1.2 **Reagents:** Potassium carbonate; 12 h, 70 °C

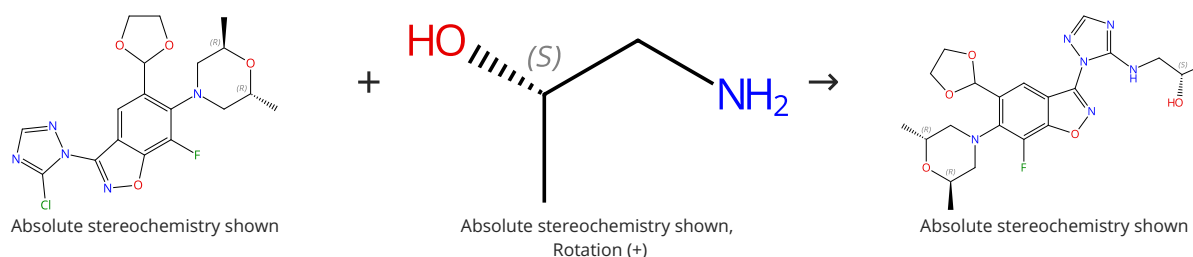
Experimental Protocols

By: Skonieczny, Kamil; et al

Angewandte Chemie, International Edition (2020), 59(37), 16104-16113.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown,
Rotation (+)

Absolute stereochemistry shown

Suppliers (81)

31-614-CAS-24414160

Steps: 1 Yield: 100%

Antibacterial Spiropyrimidinetriones with N-Linked Azole Substituents on a Benzisoxazole Scaffold Targeting DNA Gyrase

1.1 **Reagents:** Triethylamine
Solvents: Tetrahydrofuran; 6 h, 70 °C; 70 °C → rt

1.2 **Reagents:** Ammonium chloride
Solvents: Water; rt

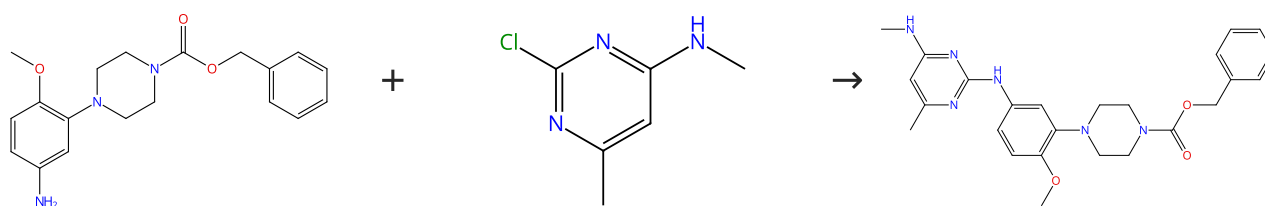
Experimental Protocols

By: Basarab, Gregory S.; et al

Journal of Medicinal Chemistry (2020), 63(20), 11882-11901.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (61)

31-172-CAS-22668740

Steps: 1 Yield: 100%

Discovery of novel histone lysine methyltransferase G9a/GLP (EHMT2/1) inhibitors: Design, synthesis, and structure-activity relationships of 2,4-diamino-6-methylpyrimidines

By: Katayama, Katsushi; et al

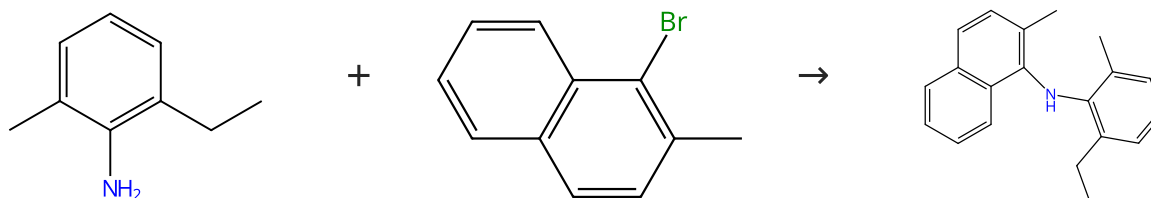
Bioorganic & Medicinal Chemistry Letters (2020), 30(20), 127475.

1.1 Reagents: Trifluoroacetic acid
Solvents: Isopropanol; rt; 1 h, 140 °C

Experimental Protocols

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

Suppliers (68)

31-172-CAS-23291187

Steps: 1 Yield: 100%

Atropisomerism in diarylamines: structural requirements and mechanisms of conformational interconversion

By: Costil, Romain; et al

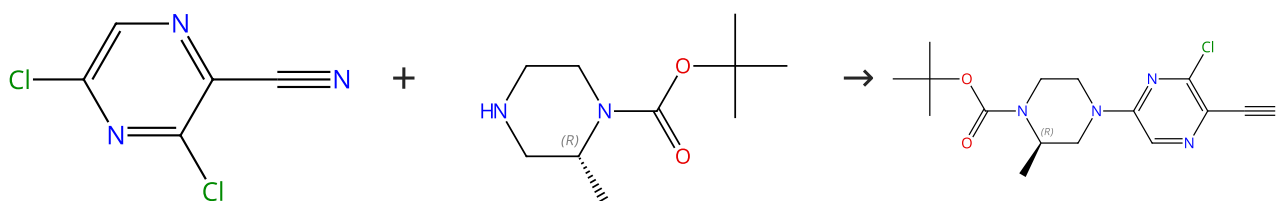
Angewandte Chemie, International Edition (2020), 59(42), 18670-18678.

1.1 Reagents: Sodium *tert*-butoxide
Catalysts: Tris(dibenzylideneacetone)dipalladium, Tris(2,2-dimethylpropyl)phosphine
Solvents: Toluene; 16 h, 80 °C

Experimental Protocols

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

 Absolute stereochemistry shown,
Rotation (-)

Suppliers (93)

Absolute stereochemistry shown

Suppliers (2)

31-172-CAS-23125154

Steps: 1 Yield: 100%

Discovery and optimization of heteroaryl piperazines as potent and selective PI3Kδ inhibitors

By: Zhou, Hua; et al

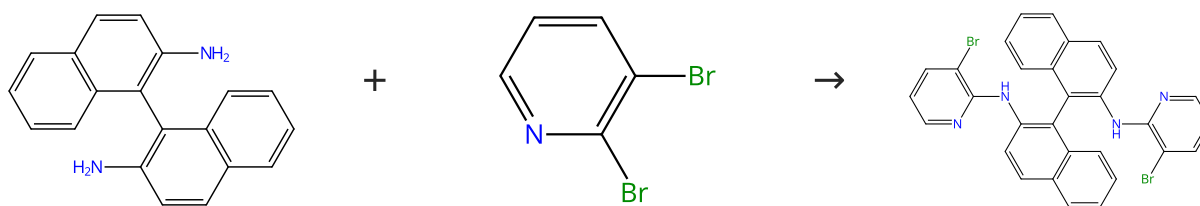
Bioorganic & Medicinal Chemistry Letters (2020), 30(1), 126715.

1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; overnight, rt

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

Suppliers (101)

31-172-CAS-23199376

Steps: 1 Yield: 100%

Synthesis and Optical Properties of Axially Chiral Bibenzo [b] carbazole Derivatives

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: *o*-Xylene; 4 h, 120 °C

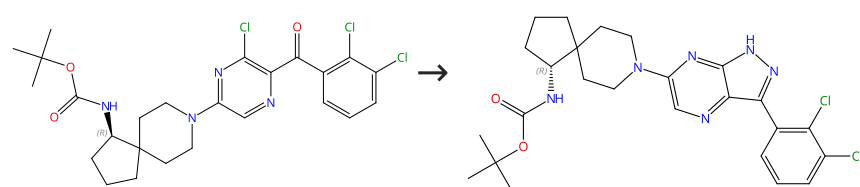
By: Takishima, Ryo; et al

Organic Letters (2021), 23(4), 1349-1354.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-614-CAS-32086541

Steps: 1 Yield: 100%

Discovery of 6-[(3S,4S)-4-Amino-3-methyl-2-oxa-8-azaspiro [4.5]decan-8-yl]-3-(2,3-dichlorophenyl)-2-methyl-3,4-dihydropyrimidin-4-one (IACS-15414), a Potent and Orally Bioavailable SHP2 Inhibitor

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Ethanol; 2 h, reflux

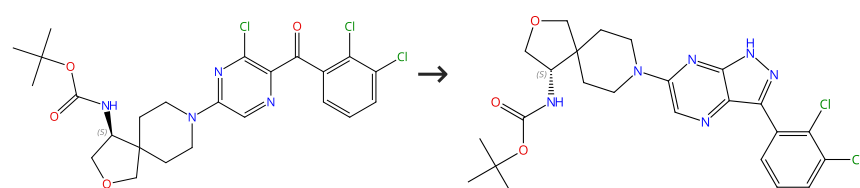
By: Czako, Barbara; et al

Journal of Medicinal Chemistry (2021), 64(20), 15141-15169.

Experimental Protocols

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

31-614-CAS-32086536

Steps: 1 Yield: 100%

Discovery of 6-[(3S,4S)-4-Amino-3-methyl-2-oxa-8-azaspiro [4.5]decan-8-yl]-3-(2,3-dichlorophenyl)-2-methyl-3,4-dihydropyrimidin-4-one (IACS-15414), a Potent and Orally Bioavailable SHP2 Inhibitor

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Ethanol; 2 h, reflux

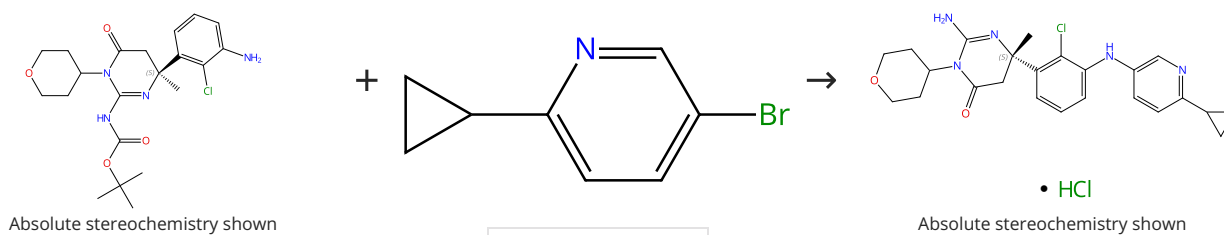
By: Czako, Barbara; et al

Journal of Medicinal Chemistry (2021), 64(20), 15141-15169.

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Absolute stereochemistry shown

31-614-CAS-39975963

Steps: 1 Yield: 100%

Discovery and Characterization of Potent, Efficacious, Orally Available Antimalarial Plasmeprin X Inhibitors and Preclinical Safety Assessment of UCB7362

By: Lowe, Martin A.; et al

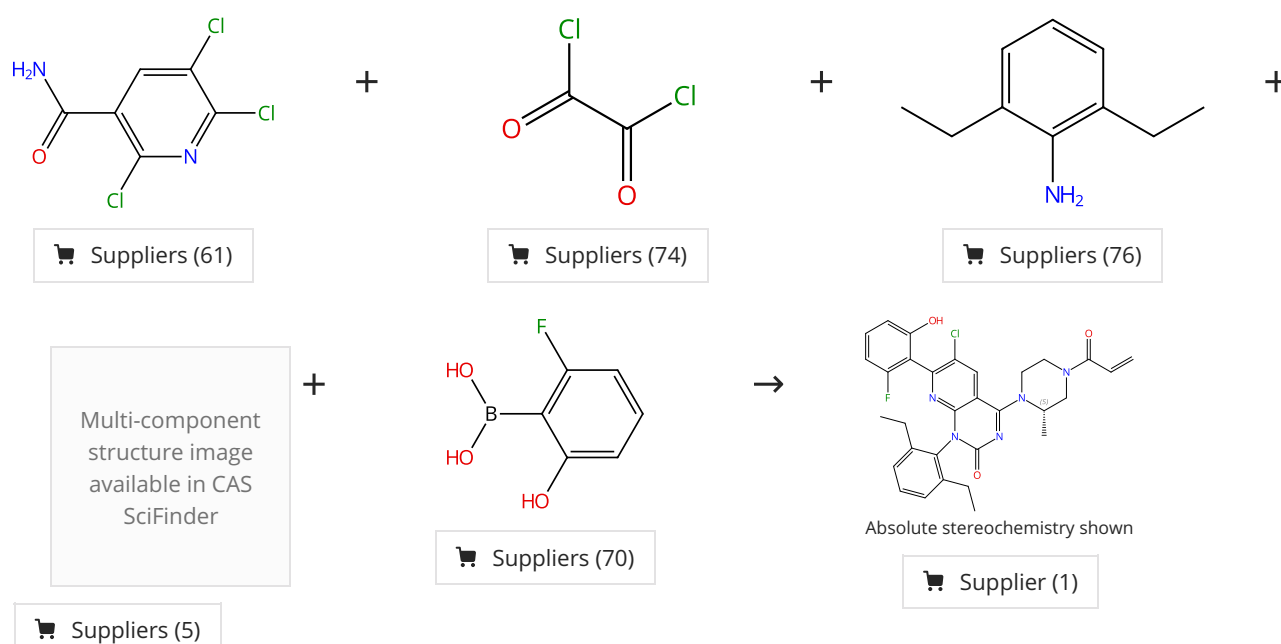
Journal of Medicinal Chemistry (2022), 65(20), 14121-14143.

- 1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Dicyclohexyl[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine, [2'-(Amino- κM)[1,1'-biphenyl]-2-yl- κC][dicyclohexyl[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine- κP](methanesulfonato- κO)palladium
Solvents: 1,4-Dioxane; 15 min, rt; 1.5 h, rt $\rightarrow$ 60 °C; 60 °C $\rightarrow$ rt; 17 h, rt
- 1.2 **Reagents:** Ammonium chloride
Solvents: Ethyl acetate, Water; rt
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Dichloromethane, 1,4-Dioxane; 2.5 h, rt

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (61)

Suppliers (74)

Suppliers (76)

Multi-component
structure image
available in CAS
SciFinder

Suppliers (70)

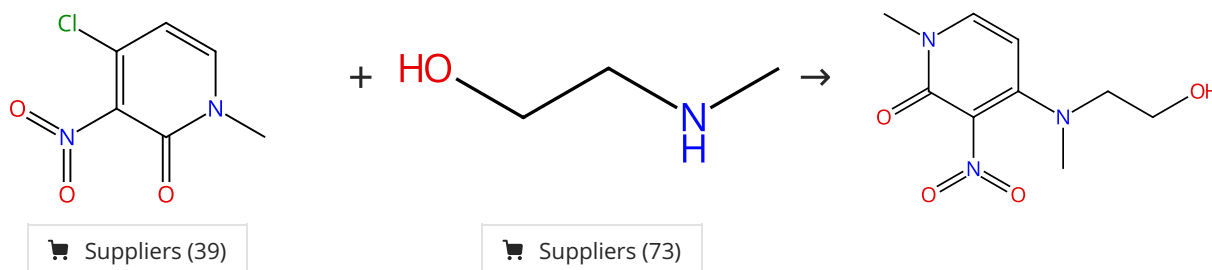
Suppliers (5)

Supplier (1)

31-172-CAS-21491832 Steps: 1 Yield: 100% 1.1 Solvents: Tetrahydrofuran; 15 min, 60 °C; 60 °C → 0 °C 1.2 Solvents: Tetrahydrofuran; 35 min, 0 °C 1.3 Reagents: Potassium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 40 min, 0 °C 1.4 Reagents: Ammonium chloride Solvents: Water 1.5 Reagents: Diisopropylethylamine, Phosphorus oxychloride Solvents: Acetonitrile; 30 min, 80 °C; 80 °C → 0 °C 1.6 Reagents: Diisopropylethylamine; 40 min, 0 °C 1.7 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane; 5 min, 90 °C 1.8 30 min, 90 °C	Discovery of a Covalent Inhibitor of K RAS^{G12C} (AMG 510) for the Treatment of Solid Tumors By: Lanman, Brian A.; et al Journal of Medicinal Chemistry (2020), 63(1), 52-65.
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Scheme 15 (1 Reaction)

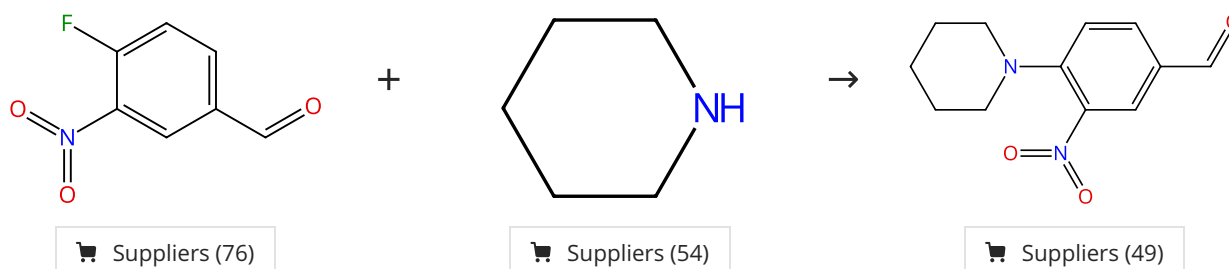
Steps: 1 Yield: 100%



31-614-CAS-35932721 Steps: 1 Yield: 100% 1.1 Solvents: Methanol; 1 h, rt → 120 °C Experimental Protocols	Mapping Ligand Interactions of Bromodomains BRD4 and ATAD2 with FragLites and PepLites-Halogenated Probes of Druglike and Peptide-like Molecular Interactions By: Davison, Gemma; et al Journal of Medicinal Chemistry (2022), 65(22), 15416-15432.
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Scheme 16 (1 Reaction)

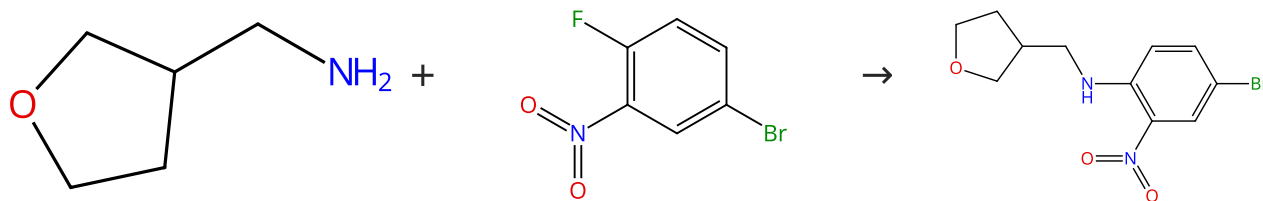
Steps: 1 Yield: 100%



31-614-CAS-32019646 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 h, 25 °C; 25 °C → rt 1.2 Solvents: Water; 30 min, rt Experimental Protocols	Expansion of the 4-(Diethylamino)benzaldehyde Scaffold to Explore the Impact on Aldehyde Dehydrogenase Activity and Antiproliferative Activity in Prostate Cancer By: Ibrahim, Ali I. M.; et al Journal of Medicinal Chemistry (2022), 65(5), 3833-3848.
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Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (90)

Suppliers (4)

31-614-CAS-34289411

Steps: 1 Yield: 100%

Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase

By: Kozlova, Arina; et al

Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

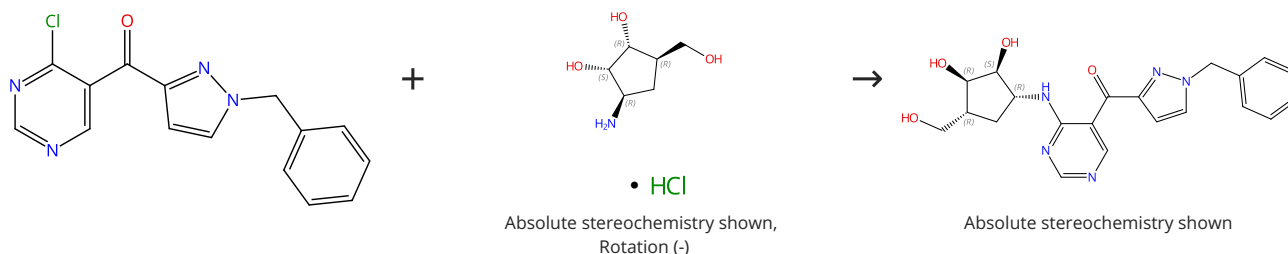
1.1 Reagents: Triethylamine

Solvents: Ethanol; 30 - 960 min, reflux

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (61)

31-614-CAS-24634839

Steps: 1 Yield: 100%

Discovery of TAK-981, a First-in-Class Inhibitor of SUMO-Activating Enzyme for the Treatment of Cancer

By: Langston, Steven P.; et al

Journal of Medicinal Chemistry (2021), 64(5), 2501-2520.

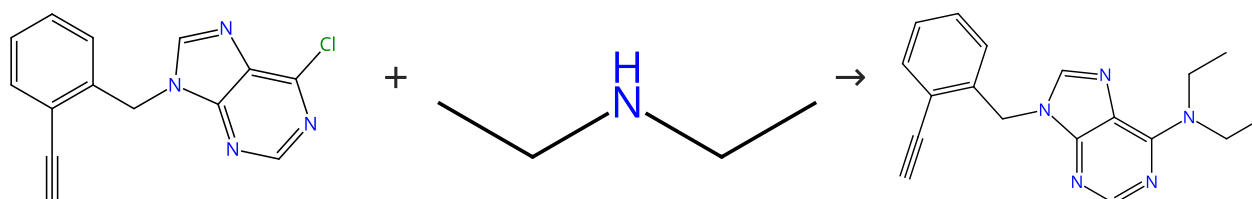
1.1 Reagents: Diisopropylethylamine

Solvents: Isopropanol; rt; 18 h, 50 °C

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (77)

31-614-CAS-38343747

Steps: 1 Yield: 100%

Discovery of a Potent Adenine-Benzyltriazolo-Pleuromutilin Conjugate with Pronounced Antibacterial Activity against MRS A

By: Heidtmann, Christoffer V.; et al

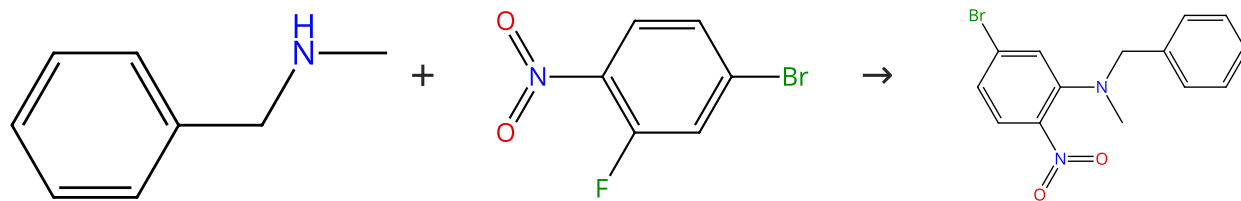
Journal of Medicinal Chemistry (2020), 63(24), 15693-15708.

1.1 Solvents: Ethanol; 4 h, 75 °C

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (98)

Suppliers (2)

31-614-CAS-31946026

Steps: 1 Yield: 100%

Structure-activity relationships of 1,4-bis(arylsulfonamido)-benzene or naphthalene-N,N'-diacetic acids with varying C2-substituents as inhibitors of Keap1-Nrf2 protein-protein interaction

By: Lee, Sumi; et al

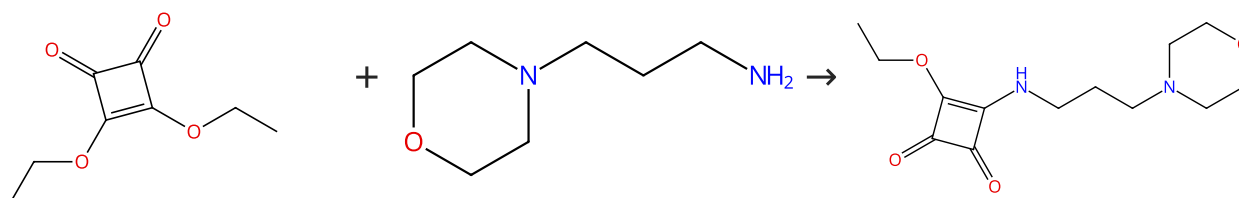
European Journal of Medicinal Chemistry (2022), 237, 114380.

- 1.1 Reagents: Sodium hydride
Solvents: Tetrahydrofuran; 5 min, 0 °C
- 1.2 0 °C; 6 h, rt

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (82)

Suppliers (89)

Supplier (1)

31-172-CAS-23209020

Steps: 1 Yield: 100%

Disrupting the Conserved Salt Bridge in the Trimerization of Influenza A Nucleoprotein

By: Woodring, Jennifer L.; et al

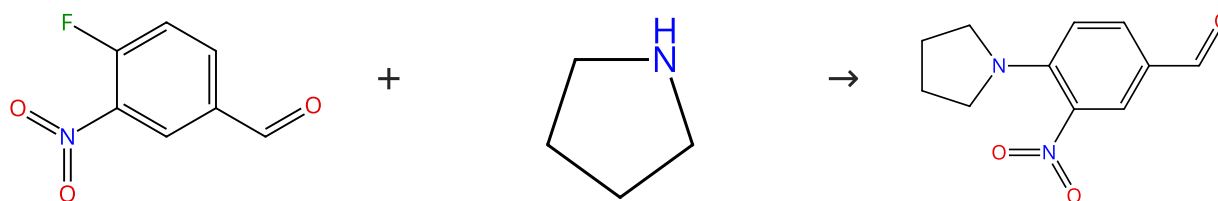
Journal of Medicinal Chemistry (2020), 63(1), 205-215.

- 1.1 Solvents: Tetrahydrofuran; 30 min, -78 °C; 18 - 30 h, rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (76)

Suppliers (75)

Suppliers (49)

31-614-CAS-32019639

Steps: 1 Yield: 100%

Expansion of the 4-(Diethylamino)benzaldehyde Scaffold to Explore the Impact on Aldehyde Dehydrogenase Activity and Antiproliferative Activity in Prostate Cancer

By: Ibrahim, Ali I. M.; et al

Journal of Medicinal Chemistry (2022), 65(5), 3833-3848.

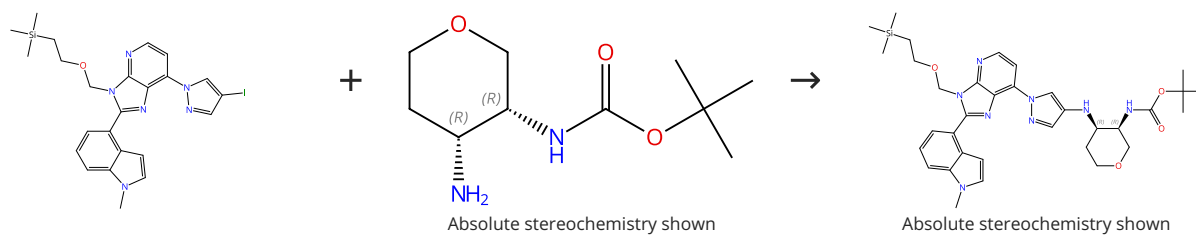
- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 1 h, 25 °C; 25 °C → rt

- 1.2 Solvents: Water; 30 min, rt

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (52)

31-614-CAS-39739241

Steps: 1 Yield: 100%

Identification of a novel series of azabenzimidazole-derived inhibitors of spleen tyrosine kinase

By: Kawatkar, Sameer P.; et al

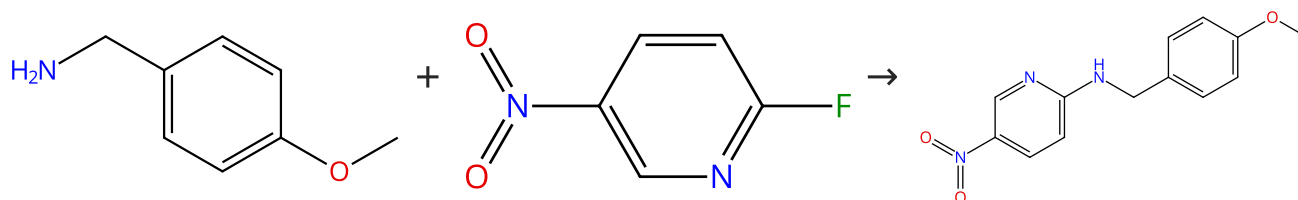
Bioorganic & Medicinal Chemistry Letters (2020), 30(18), 127393.

1.1 **Reagents:** Potassium carbonate
Catalysts: L-Proline, Cuprous iodide
Solvents: Dimethyl sulfoxide; 16 h, 100 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

Suppliers (84)

Suppliers (11)

31-614-CAS-24903865

Steps: 1 Yield: 100%

Structure-Guided Development of Small-Molecule PRC2 Inhibitors Targeting EZH2-EED Interaction

By: Du, Daohai; et al

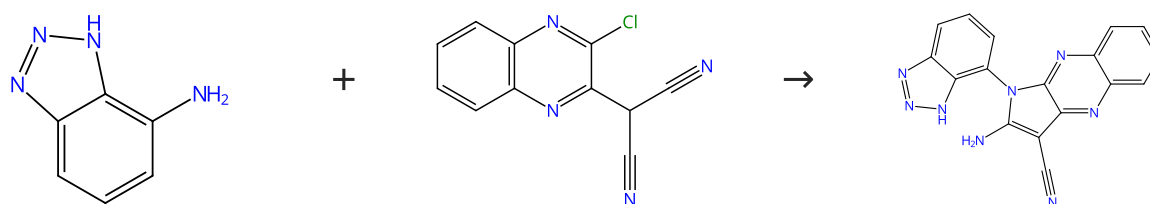
Journal of Medicinal Chemistry (2021), 64(12), 8194-8207.

1.1 **Reagents:** Sodium carbonate
Solvents: Dimethylformamide; 4 h, 50 °C

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (51)

Suppliers (13)

31-614-CAS-40132722

Steps: 1 Yield: 100%

Discovery of an Orally Bioavailable and Selective PKMYT1 Inhibitor, RP-6306

By: Szychowski, Janek; et al

Journal of Medicinal Chemistry (2022), 65(15), 10251-10284.

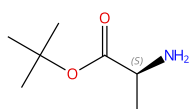
1.1 **Solvents:** *N*-Methyl-2-pyrrolidone; 1 h, 150 °C; 150 °C → rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; rt

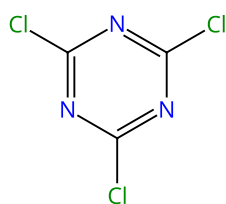
Experimental Protocols

Scheme 26 (1 Reaction)

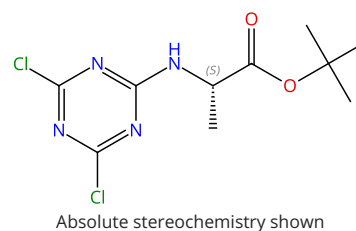
Steps: 1 Yield: 100%



+



→



• HCl

Absolute stereochemistry shown,
Rotation (+)

Suppliers (71)

Suppliers (87)

31-614-CAS-24247002

Steps: 1 Yield: 100%

Design, synthesis, and functional evaluation of triazine-based bivalent agents that simultaneously target the active site and hot spot of phosphatase Cdc25B

1.1 Reagents: Diisopropylethylamine
Solvents: Dichloromethane; 2 h, rt

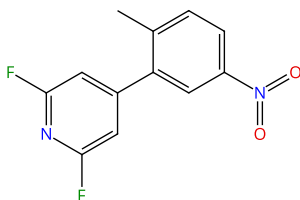
Experimental Protocols

By: Nagaoka, Yosei; et al

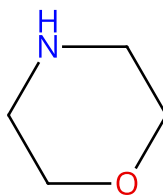
Bioorganic & Medicinal Chemistry Letters (2021), 48, 128265.

Scheme 27 (1 Reaction)

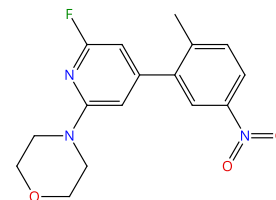
Steps: 1 Yield: 100%



+



→



Suppliers (88)

31-172-CAS-21642305

Steps: 1 Yield: 100%

Design and Discovery of N-(3-(2-(2-Hydroxyethoxy)-6-morpholinopyridin-4-yl)-4-methylphenyl)-2-(trifluoromethyl)isonicotinamide, a Selective, Efficacious, and Well-Tolerated RAS AF Inhibitor Targeting RAS Mutant Cancers: The Path to the Clinic

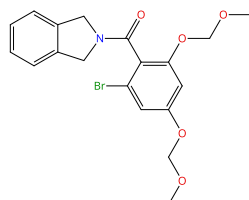
1.1 Reagents: Diisopropylethylamine
Solvents: Ethanol; rt; 18 h, 55 °C

By: Ramurthy, Savithri; et al

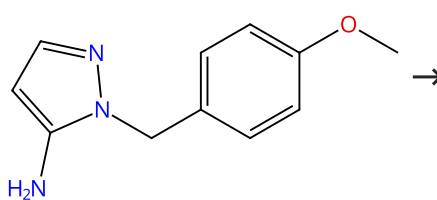
Journal of Medicinal Chemistry (2020), 63(5), 2013-2027.

Scheme 28 (1 Reaction)

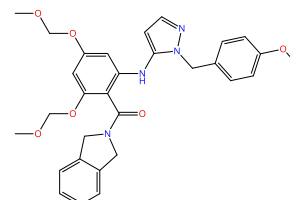
Steps: 1 Yield: 100%



+



→



Supplier (1)

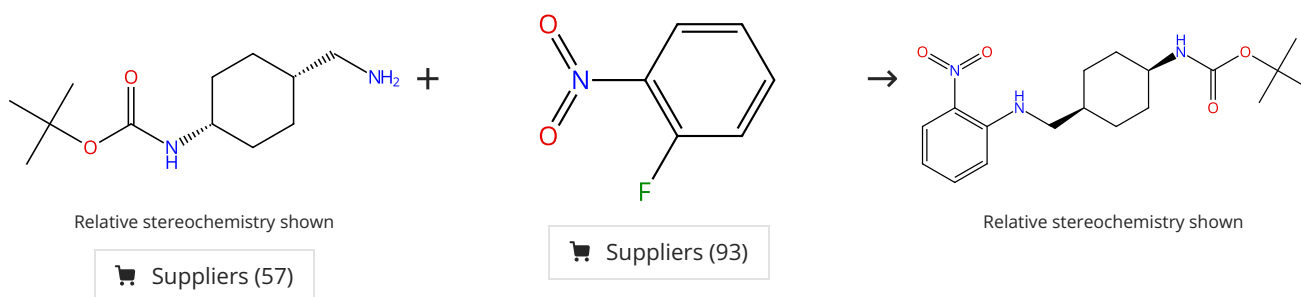
Suppliers (71)

Supplier (1)

31-172-CAS-21608400 Steps: 1 Yield: 100% 1.1 Reagents: Sodium phenoxide Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; 2 h, 120 °C; 120 °C → rt 1.2 Reagents: Trithiocyanuric acid Solvents: Toluene; overnight, rt Experimental Protocols	Design and Synthesis of Fungal-Selective Resorcyate Aminopyrazole Hsp90 Inhibitors By: Huang, David S.; et al Journal of Medicinal Chemistry (2020), 63(5), 2139-2180.
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Scheme 29 (1 Reaction)

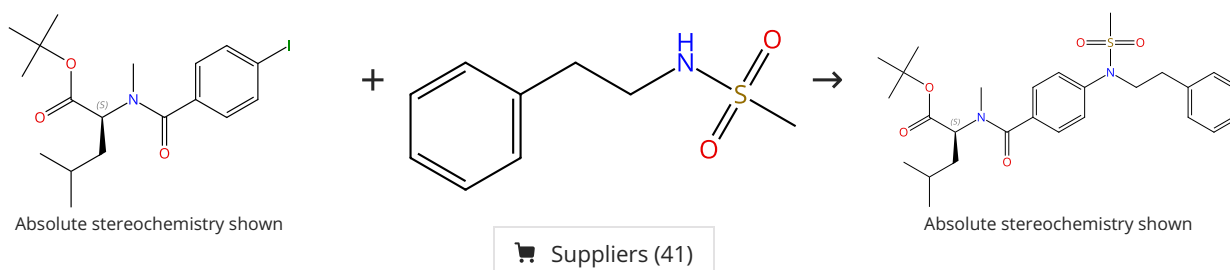
Steps: 1 Yield: 100%



31-172-CAS-21630679 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; overnight, 100 °C	Discovery of a Bromodomain and Extraterminal Inhibitor with a Low Predicted Human Dose through Synergistic Use of Encoded Library Technology and Fragment Screening By: Wellaway, Christopher R.; et al Journal of Medicinal Chemistry (2020), 63(2), 714-746.
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Scheme 30 (1 Reaction)

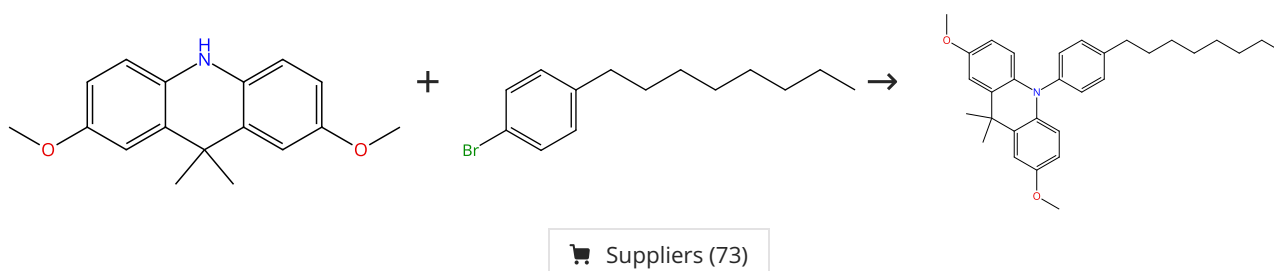
Steps: 1 Yield: 100%



31-614-CAS-24353260 Steps: 1 Yield: 100% 1.1 Reagents: Tripotassium phosphate Catalysts: <i>trans</i>-1,2-Diaminocyclohexane, Cuprous iodide Solvents: 1,4-Dioxane; 36 h, 110 °C 1.2 Reagents: Sodium hydroxide, Ammonia Solvents: Water Experimental Protocols	Modulators of hERAP2 discovered by high-throughput screening By: Laura, Medve; et al European Journal of Medicinal Chemistry (2021), 211, 113053.
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Scheme 31 (1 Reaction)

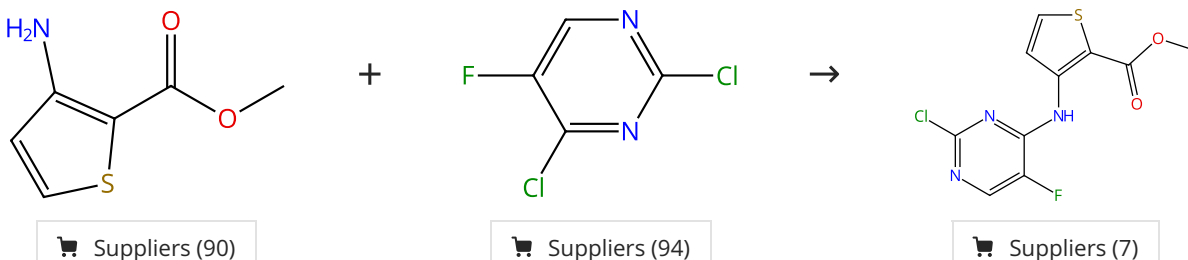
Steps: 1 Yield: 100%



31-172-CAS-23070673	Steps: 1 Yield: 100%	An Electroactive Pure Organic Room-Temperature Phosphorescence Polymer Based on a Donor-Oxygen-Acceptor Geometry
1.1 Reagents: Sodium <i>tert</i> -butoxide Catalysts: Tris(dibenzylideneacetone)dipalladium, Tri- <i>tert</i> -butylphosphonium tetrafluoroborate Solvents: Toluene; 105 °C		By: Liu, Xinrui; et al
Experimental Protocols		Angewandte Chemie, International Edition (2021), 60(5), 2455-2463.

Scheme 32 (1 Reaction)

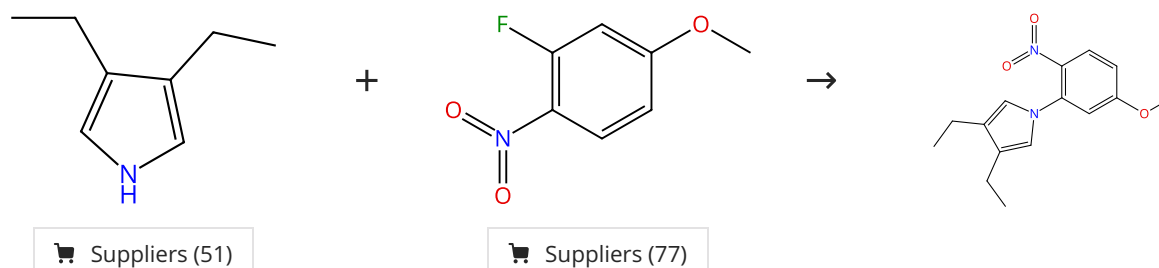
Steps: 1 Yield: 100%



31-172-CAS-23338551	Steps: 1 Yield: 100%	Discovery of methyl 3-((2-((1-(dimethylglycyl)-5-methoxyindolin-6-yl)amino)-5-(trifluoro-methyl) pyrimidin-4-yl)amino) thiophene-2-carboxylate as a potent and selective polo-like kinase 1 (PLK1) inhibitor for combating hepatocellular carcinoma
1.1 Reagents: Sodium hydride Solvents: Dimethylformamide; 15 min, 0 °C		By: Deng, Zhou; et al
1.2 0 °C		European Journal of Medicinal Chemistry (2020), 206, 112697.
Experimental Protocols		

Scheme 33 (1 Reaction)

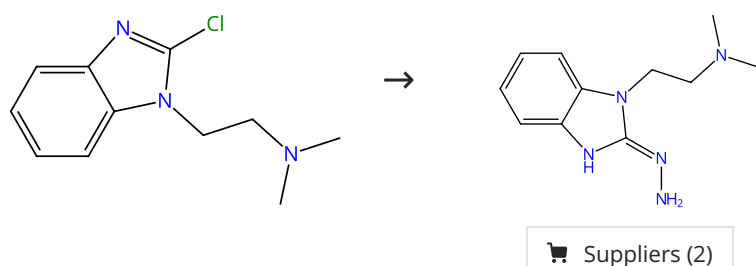
Steps: 1 Yield: 100%



31-172-CAS-23257891	Steps: 1 Yield: 100%	Enantioselective Construction of Quinoxaline-Based Heterobiaryls and P,N-Ligands via Chirality Transfer Strategy
1.1 Reagents: Sodium hydroxide Solvents: Dimethyl sulfoxide; 2 h, rt		By: Gao, Zeng; et al
Experimental Protocols		Organic Letters (2021), 23(4), 1181-1187.

Scheme 34 (1 Reaction)

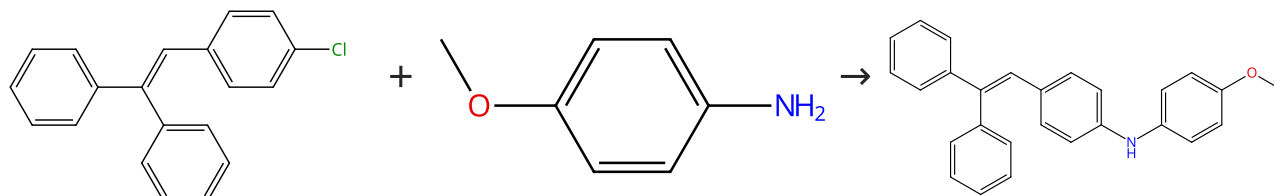
Steps: 1 Yield: 100%



31-614-CAS-31542248	Steps: 1 Yield: 100%	Benzothiazolyl and Benzoxazolyl Hydrazones Function as Zinc Metallochaperones to Reactivate Mutant p53
1.1 Reagents: Hydrazine hydrate (1:1) Solvents: Isopropanol; overnight, 140 °C		By: Gilleran, John A.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(4), 2024-2045.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (6)

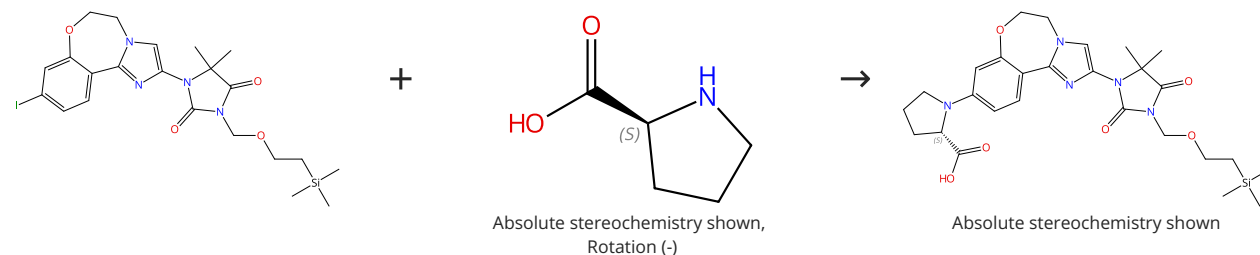
Suppliers (83)

Suppliers (3)

31-614-CAS-30040122	Steps: 1 Yield: 100%	Pd-Catalyzed Coupling of N-Tosylhydrazones with Benzylic Phosphates: Toward the Synthesis of Di- or Tri-Substituted Alkenes
1.1 Reagents: Potassium <i>tert</i> -butoxide Catalysts: Di- μ -chlorobis(η^3 -2-propenyl)dipalladium, 1 <i>H</i> -Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-4,5-dihydro-, chloride (1:1) Solvents: 1,4-Dioxane; 1.5 h, 100 °C		By: Zhang, Kena; et al
Experimental Protocols		Journal of Organic Chemistry (2022), 87(2), 1249-1261.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)

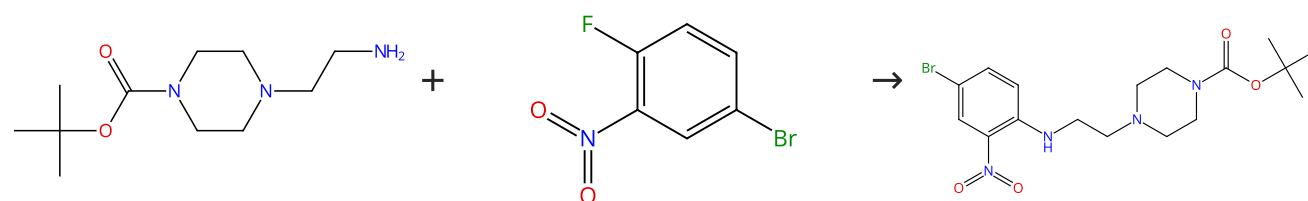
Absolute stereochemistry shown

Suppliers (169)

31-614-CAS-35032470	Steps: 1 Yield: 100%	Discovery of GDC-0077 (Inavolisib), a Highly Selective Inhibitor and Degradar of Mutant PI3Ka
1.1 Reagents: Tripotassium phosphate Catalysts: Cuprous iodide Solvents: Dimethyl sulfoxide; 16 h, 120 °C		By: Hanan, Emily J.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2022), 65(24), 16589-16621.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



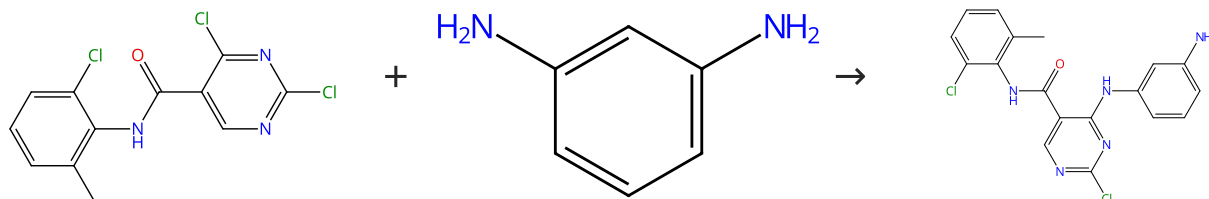
Suppliers (90)

Suppliers (90)

31-614-CAS-34289445	Steps: 1 Yield: 100%	Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase By: Kozlova, Arina; et al Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.
1.1 Reagents: Triethylamine Solvents: Ethanol; 2.5 h, reflux		
Experimental Protocols		

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



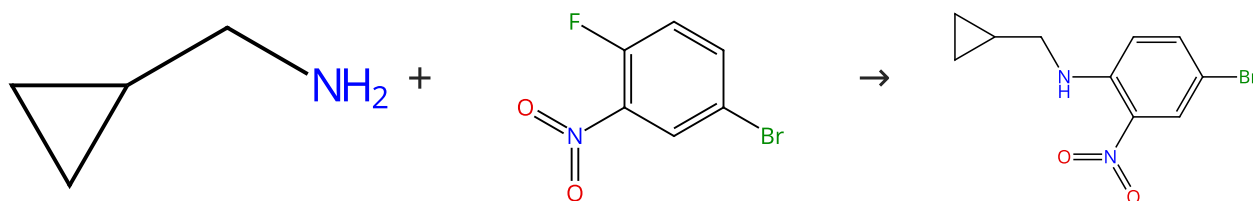
Suppliers (2)

Suppliers (75)

31-614-CAS-23938110	Steps: 1 Yield: 100%	Structure-Based Design of a Potent and Selective Covalent Inhibitor for SRC Kinase That Targets a P-Loop Cysteine By: Du, Guangyan; et al Journal of Medicinal Chemistry (2020), 63(4), 1624-1641.
1.1 Reagents: Diisopropylethylamine Solvents: 1-Butanol; 1 h, rt → 90 °C		
Experimental Protocols		

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

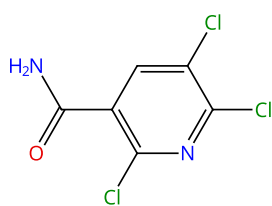
Suppliers (90)

Suppliers (39)

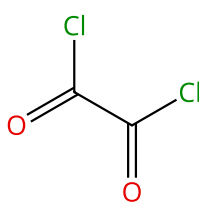
31-614-CAS-34289428	Steps: 1 Yield: 100%	Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase By: Kozlova, Arina; et al Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.
1.1 Reagents: Triethylamine Solvents: Ethanol; 30 - 960 min, reflux		
Experimental Protocols		

Scheme 40 (1 Reaction)

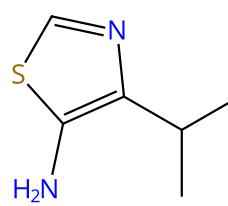
Steps: 1 Yield: 100%



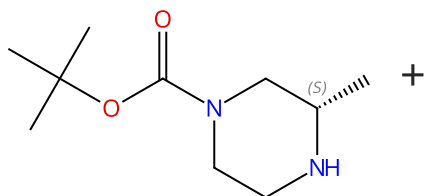
Suppliers (61)



Suppliers (74)

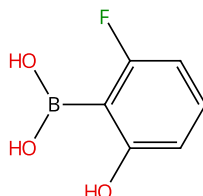


Suppliers (35)

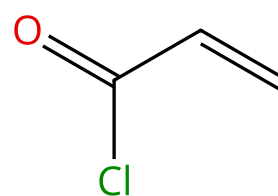


Absolute stereochemistry shown

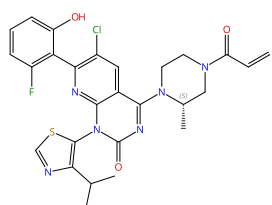
Suppliers (100)



Suppliers (70)



Suppliers (41)



Absolute stereochemistry shown

Supplier (1)

31-172-CAS-21491829

Steps: 1 Yield: 100%

- 1.1 **Solvents:** 1,2-Dichloroethane; rt; 30 min, 80 °C
- 1.2 **Solvents:** Acetonitrile; 15 min, rt
- 1.3 **Reagents:** Potassium bis(trimethylsilyl)amide
Solvents: Tetrahydrofuran; 10 min, 0 °C; 30 min, rt
- 1.4 **Reagents:** Ammonium chloride
Solvents: Water; rt
- 1.5 **Reagents:** Diisopropylethylamine, Phosphorus oxychloride
Solvents: Acetonitrile; rt; 30 min, 80 °C
- 1.6 **Reagents:** Diisopropylethylamine
Solvents: Dimethylformamide; 10 min, rt
- 1.7 **Reagents:** Water; 15 min, cooled
- 1.8 **Reagents:** Potassium acetate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane; 40 min, 80 °C
- 1.9 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; 30 min, rt
- 1.10 **Reagents:** Diisopropylethylamine
Solvents: Dichloromethane; 1 h, 0 °C

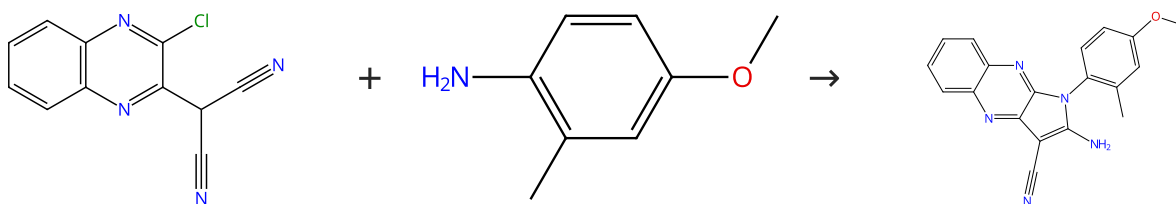
Discovery of a Covalent Inhibitor of K RAS^{G12C} (AMG 510) for the Treatment of Solid Tumors

By: Lanman, Brian A.; et al

Journal of Medicinal Chemistry (2020), 63(1), 52-65.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (13)

Suppliers (85)

31-614-CAS-40132705

Steps: 1 Yield: 100%

Discovery of an Orally Bioavailable and Selective PKMYT1 Inhibitor, RP-63061.1 Solvents: *N*-Methyl-2-pyrrolidone; 1 h, rt → 130 °C; 130 °C → rt

1.2 Reagents: Sodium bicarbonate

Solvents: Water; rt

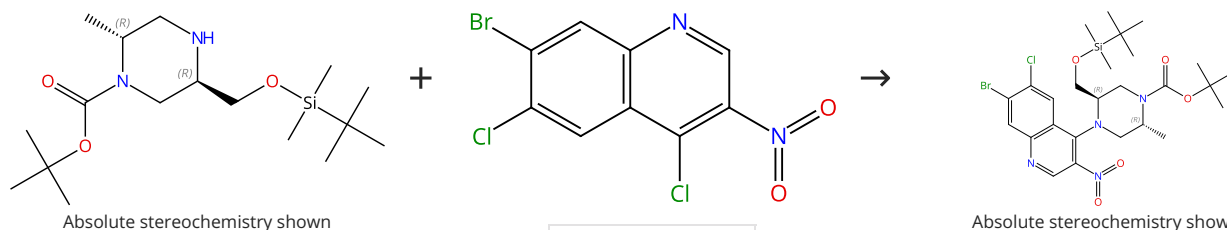
By: Szychowski, Janek; et al

Journal of Medicinal Chemistry (2022), 65(15), 10251-10284.

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Supplier (1)

Suppliers (3)

Supplier (1)

31-614-CAS-24890707

Steps: 1 Yield: 100%

Structure-Based Design and Pharmacokinetic Optimization of Covalent Allosteric Inhibitors of the Mutant GTPase KRAS^{G12C}

1.1 Reagents: Diisopropylethylamine

Solvents: Isopropanol; 2 h, 80 °C

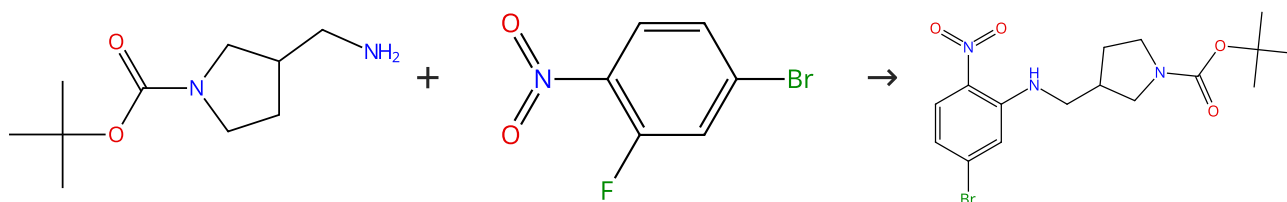
By: Kettle, Jason G.; et al

Journal of Medicinal Chemistry (2020), 63(9), 4468-4483.

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (74)

Suppliers (98)

Supplier (1)

31-614-CAS-34289481

Steps: 1 Yield: 100%

Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase

1.1 Reagents: Triethylamine

Solvents: Ethanol; 30 - 960 min, reflux

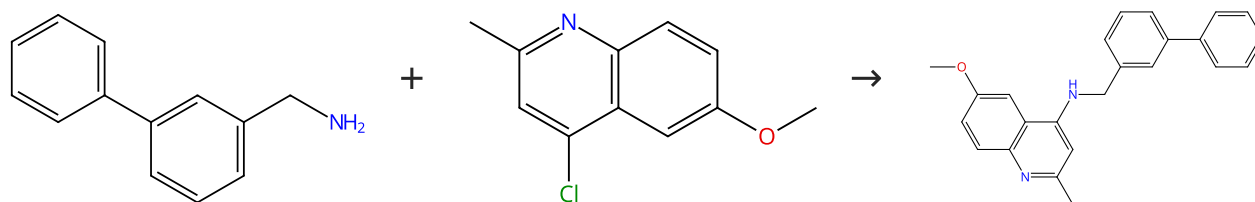
By: Kozlova, Arina; et al

Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

Experimental Protocols

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (73)

Suppliers (64)

31-614-CAS-31592593

Steps: 1 Yield: 100%

Discovery of AL-GDa62 as a Potential Synthetic Lethal Lead for the Treatment of Gastric Cancer

By: Luxenburger, Andreas; et al

Journal of Medicinal Chemistry (2021), 64(24), 18114-18142.

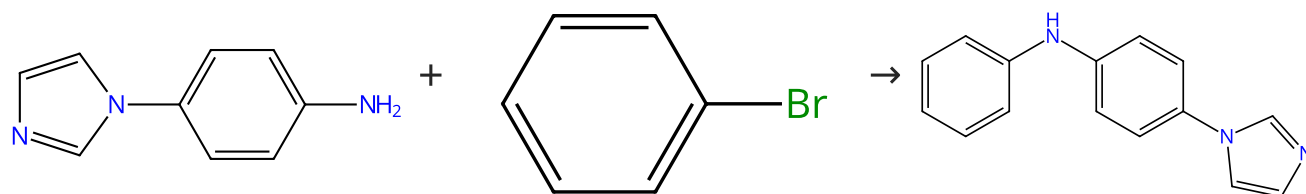
1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, BINAP
Solvents: 1,4-Dioxane; rt → 110 °C; overnight, 110 °C; 110 °C → rt

1.2 **Solvents:** Water; rt

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (95)

Suppliers (80)

31-172-CAS-22693830

Steps: 1 Yield: 100%

Development of an Aryl Amination Catalyst with Broad Scope Guided by Consideration of Catalyst Stability

By: McCann, Scott D.; et al

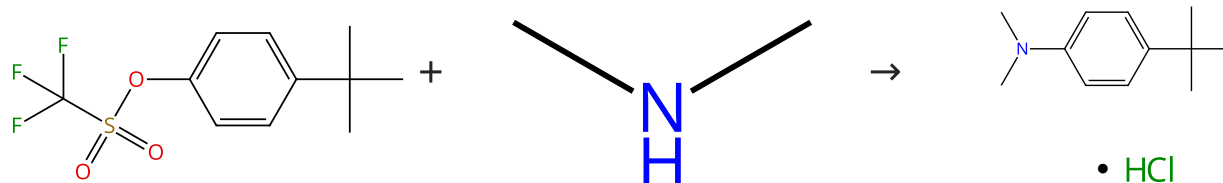
Journal of the American Chemical Society (2020), 142(35), 15027-15037.

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC¹]phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (*SP*-4-2)-
Solvents: Tetrahydrofuran, Dodecane; 1 h, 90 °C

1.2 **Reagents:** Oxygen
Solvents: Ethyl acetate

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



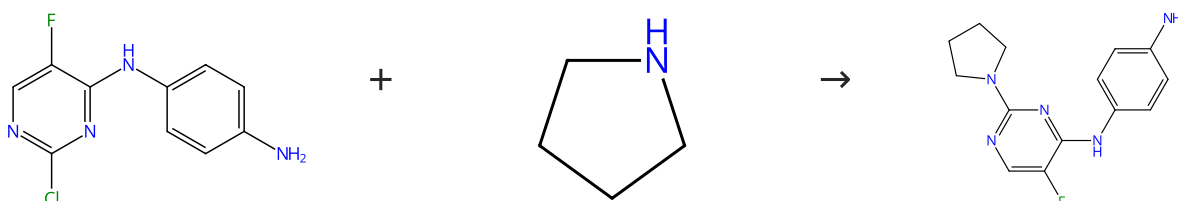
Suppliers (43)

Suppliers (118)

31-172-CAS-22808188 Steps: 1 Yield: 100% 1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: Tetrahydrofuran; 5 min, 80 °C 1.2 Solvents: Tetrahydrofuran; 16 h, 80 °C 1.3 Reagents: Water 1.4 Reagents: Hydrochloric acid Solvents: Diethyl ether	Palladium-Catalyzed Synthesis of N,N-Dimethylanilines via Buchwald-Hartwig Amination of (Hetero)aryl Triflates By: Taeufer, Tobias; et al Journal of Organic Chemistry (2020), 85(11), 7097-7111.
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Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%

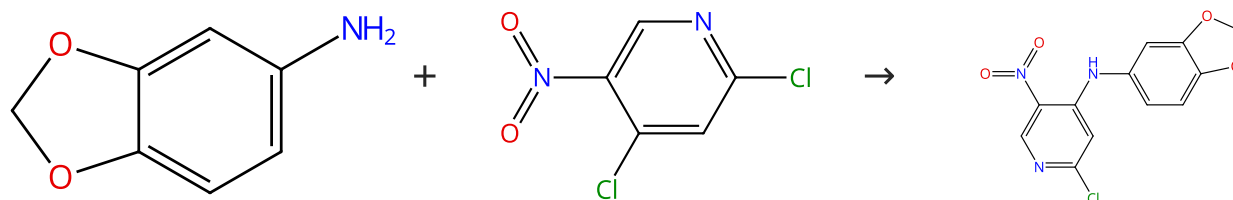


Suppliers (75)

31-614-CAS-39885320 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 80 °C	Design, Synthesis, and Biological Evaluation of the First Inhibitors of Oncogenic CHD1L By: Prigaro, Brett J.; et al Journal of Medicinal Chemistry (2022), 65(5), 3943-3961.
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Scheme 48 (1 Reaction)

Steps: 1 Yield: 100%



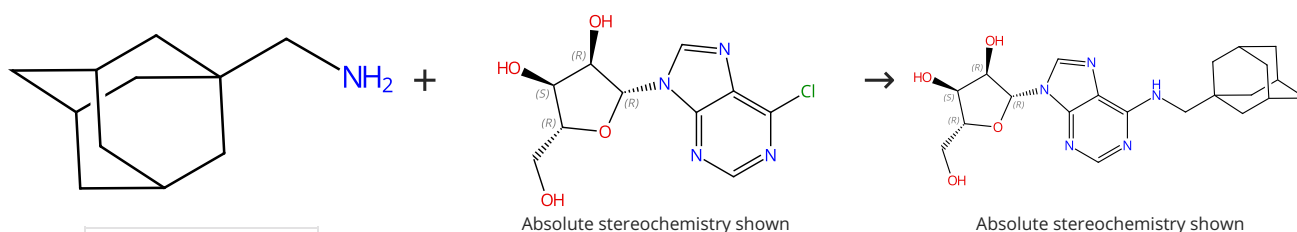
Suppliers (75)

Suppliers (90)

31-614-CAS-31332688 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine Solvents: Acetonitrile; rt Experimental Protocols	A Phenotypic Screen Identifies a Compound Series That Induces Differentiation of Acute Myeloid Leukemia Cells In Vitro and Shows Antitumor Effects In Vivo By: Josa-Cullere, Laia; et al Journal of Medicinal Chemistry (2021), 64(21), 15608-15628.
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Scheme 49 (1 Reaction)

Steps: 1 Yield: 100%



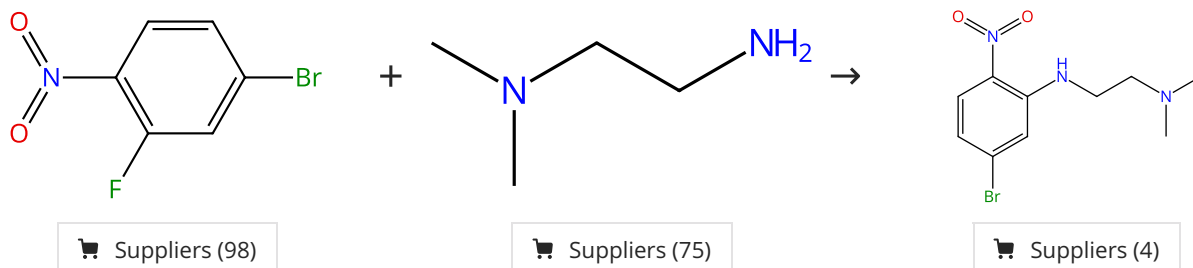
Suppliers (104)

Suppliers (88)

31-614-CAS-34772023	Steps: 1 Yield: 100%	Synthesis of Fluorescent Probes Targeting Tumor-Suppressor Protein FHIT and Identification of Apoptosis-Inducing FHIT Inhibitors
1.1 Solvents: Ethanol; 2.5 h, 60 °C		By: Kawaguchi, Mitsuyasu; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(13), 9567-9576.

Scheme 50 (1 Reaction)

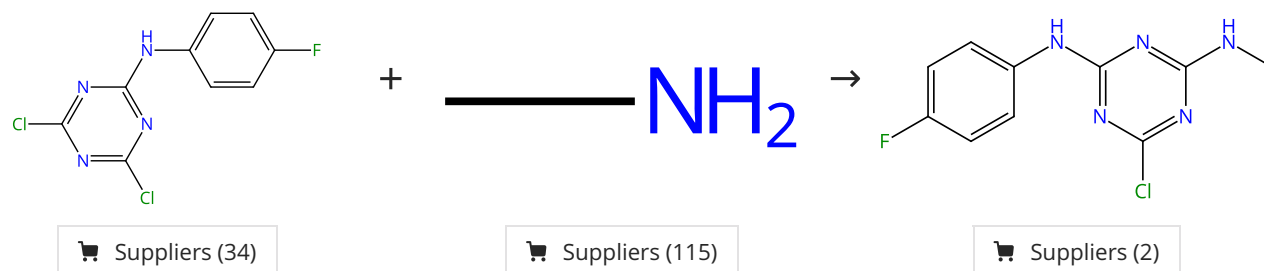
Steps: 1 Yield: 100%



31-614-CAS-34289460	Steps: 1 Yield: 100%	Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase
1.1 Reagents: Triethylamine Solvents: Ethanol; 30 - 960 min, reflux		By: Kozlova, Arina; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

Scheme 51 (1 Reaction)

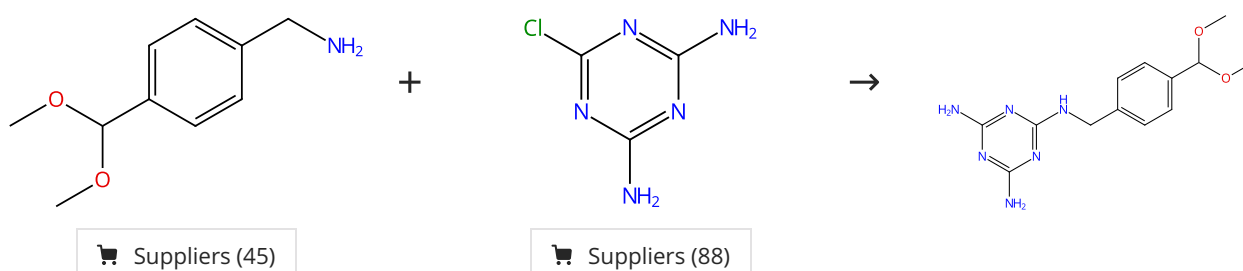
Steps: 1 Yield: 100%



31-172-CAS-23893124	Steps: 1 Yield: 100%	Discovery of Small-Molecule Cyclic GMP-AMP Synthase Inhibitors
1.1 Reagents: Triethylamine Solvents: Isopropanol; 45 °C		By: Padilla-Salinas, Rosaura; et al
		Journal of Organic Chemistry (2020), 85(3), 1579-1600.

Scheme 52 (1 Reaction)

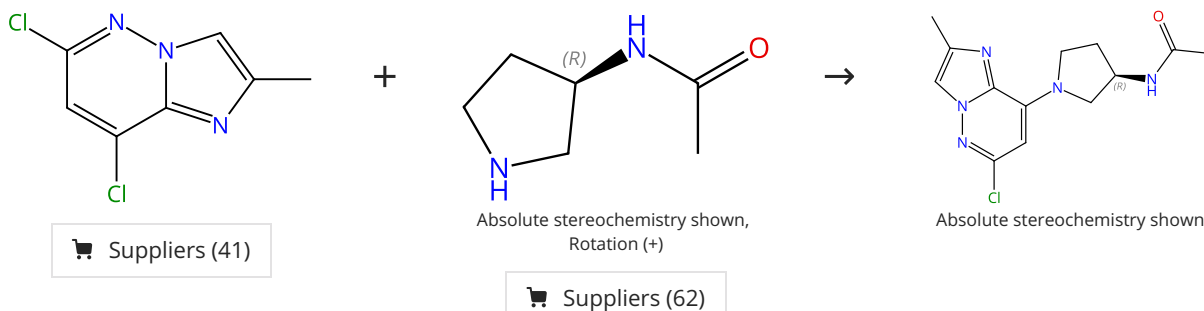
Steps: 1 Yield: 100%



31-614-CAS-40187300	Steps: 1 Yield: 100%	Dynamic Covalent Template-Guided Screen for Nucleic Acid-Targeting Agents
1.1 Solvents: Methanol, Acetonitrile; 25 °C; 1 h, 25 °C		By: Krueger, Sarah B.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2022), 65(18), 12417-12426.

Scheme 53 (1 Reaction)

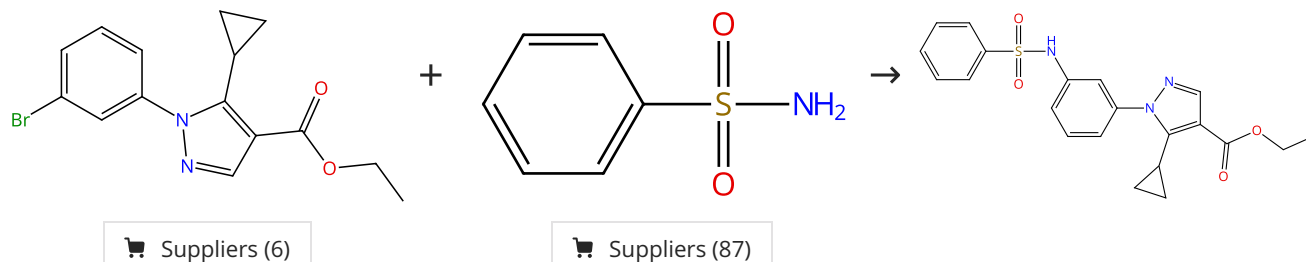
Steps: 1 Yield: 100%



31-172-CAS-22355812	Steps: 1 Yield: 100%	Novel Human Neutral Sphingomyelinase 2 Inhibitors as Potential Therapeutics for Alzheimer's Disease
1.1 Reagents: Diisopropylethylamine Solvents: Acetonitrile; 16 h, 85 °C		By: Sala, Michal; et al
		Journal of Medicinal Chemistry (2020), 63(11), 6028-6056.

Scheme 54 (1 Reaction)

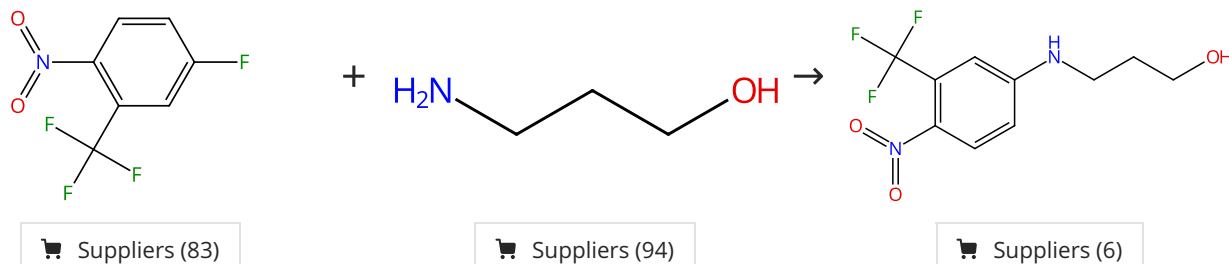
Steps: 1 Yield: 100%



31-614-CAS-25850521	Steps: 1 Yield: 100%	Deconstructing Noncovalent Kelch-like ECH-Associated Protein 1 (Keap1) Inhibitors into Fragments to Reconstruct New Potent Compounds
1.1 Reagents: Tripotassium phosphate Catalysts: <i>N,N</i> -Dimethylglycine, Cuprous iodide Solvents: Dimethylformamide; 20 h, 160 °C		By: Pallesen, Jakob S.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(8), 4623-4661.

Scheme 55 (1 Reaction)

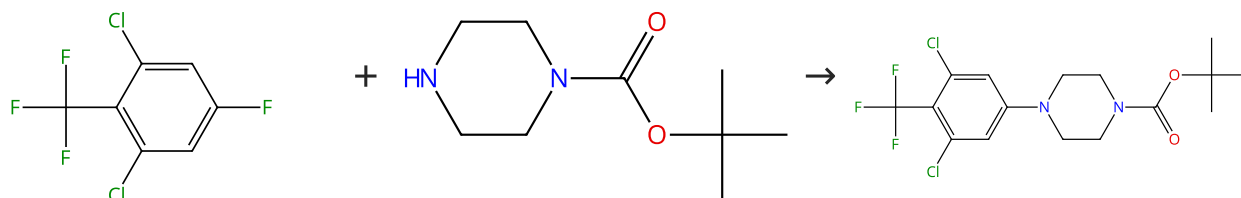
Steps: 1 Yield: 100%



31-614-CAS-34492550	Steps: 1 Yield: 100%	Nitric Oxide Photo-Donor Hybrids of Ciprofloxacin and Norfloxacin: A Shift in Activity from Antimicrobial to Anticancer Agents
1.1 Solvents: Acetonitrile; 12 h, 60 °C		By: Fallica, Antonino Nicolo; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(15), 11597-11613.

Scheme 56 (1 Reaction)

Steps: 1 Yield: 100%



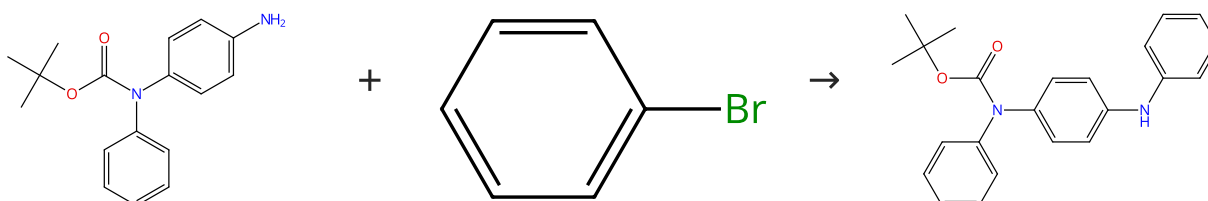
Suppliers (39)

Suppliers (110)

31-614-CAS-32750286	Steps: 1 Yield: 100%	Structural, electronic, and electrostatic determinants for inhibitor binding to subsites S1 and S2 in SARS-CoV-2 main protease
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 24 h, 100 °C		By: Kneller, Daniel W.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(23), 17366-17383.

Scheme 57 (1 Reaction)

Steps: 1 Yield: 100%



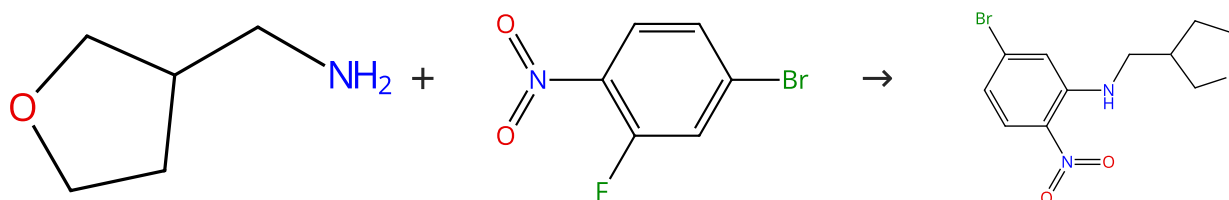
Suppliers (2)

Suppliers (80)

31-172-CAS-21543695	Steps: 1 Yield: 100%	Thermally Activated Delayed Fluorescent Donor-Acceptor-Donor-Acceptor π-Conjugated Macrocycle for Organic Light-Emitting Diodes
1.1 Reagents: Cesium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, Bis(1,1-dimethylethyl)[2',4',6'-tris(1-methylethyl)][1,1'-biphenyl]-2-yl]phosphine Solvents: <i>tert</i> -Butanol, Tetrahydrofuran; 24 h, 85 °C		By: Izumi, Saika; et al
Experimental Protocols		Journal of the American Chemical Society (2020), 142(3), 1482-1491.

Scheme 58 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

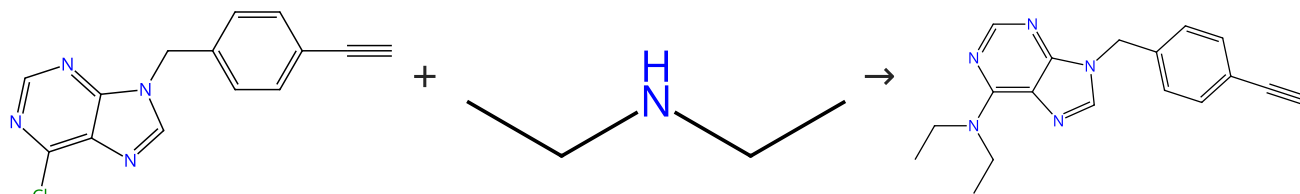
Suppliers (98)

Suppliers (4)

31-614-CAS-34289457	Steps: 1 Yield: 100%	Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase
1.1 Reagents: Triethylamine Solvents: Ethanol; 30 - 960 min, reflux		By: Kozlova, Arina; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.

Scheme 59 (1 Reaction)

Steps: 1 Yield: 100%

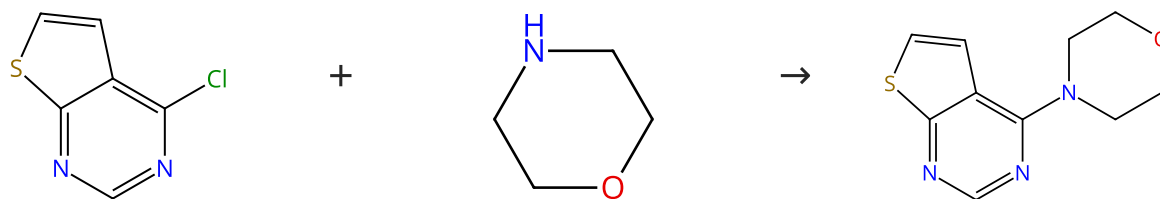


Suppliers (77)

31-614-CAS-38343758	Steps: 1 Yield: 100%	Discovery of a Potent Adenine-Benzyltriazolo-Pleuromutilin Conjugate with Pronounced Antibacterial Activity against MRS A
1.1 Solvents: Ethanol; 3.5 h, 75 °C		By: Heidtmann, Christoffer V.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2020), 63(24), 15693-15708.

Scheme 60 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

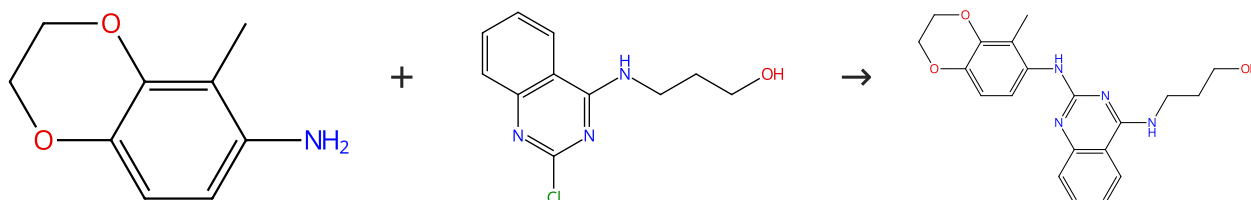
Suppliers (88)

Suppliers (7)

31-172-CAS-21551663	Steps: 1 Yield: 100%	Rapid Access to Kinase Inhibitor Pharmacophores by Regioselective C-H Arylation of Thieno[2,3-d]pyrimidines
1.1 Solvents: Dimethylformamide; 4 h, 110 °C		By: Yamada, Shuya; et al
Experimental Protocols		Organic Letters (2020), 22(4), 1547-1551.

Scheme 61 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (39)

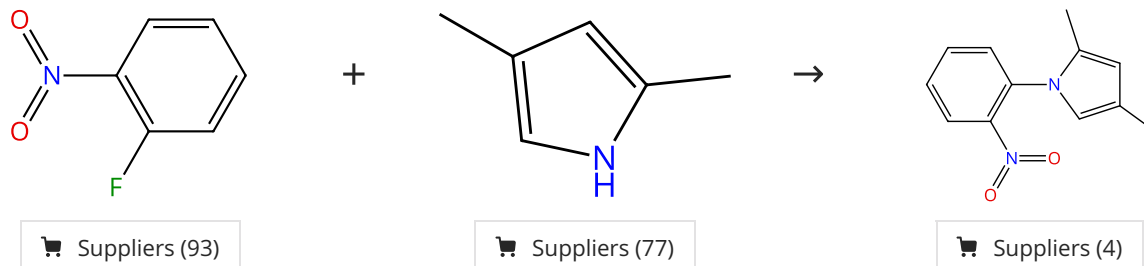
Suppliers (16)

Supplier (1)

31-614-CAS-40275654 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol; 30 - 60 min, 150 °C Experimental Protocols	A Druglike Small Molecule that Targets r(CCUG) Repeats in Myotonic Dystrophy Type 2 Facilitates Degradation by RNA Quality Control Pathways By: Wagner-Griffin, Sarah; et al Journal of Medicinal Chemistry (2021), 64(12), 8474-8485.
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Scheme 62 (1 Reaction)

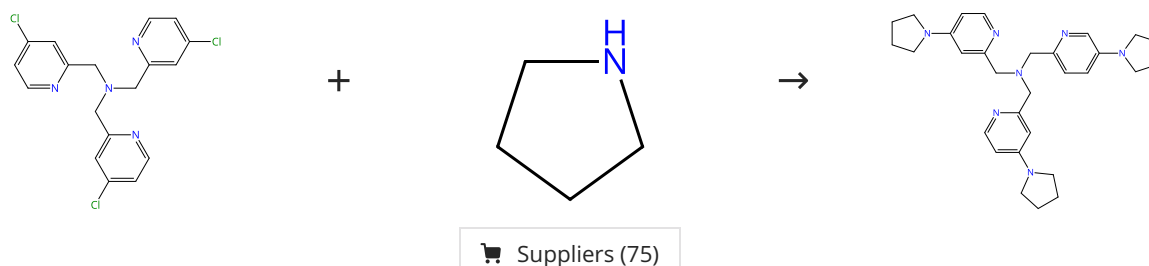
Steps: 1 Yield: 100%



31-172-CAS-23258221 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydroxide Solvents: Dimethyl sulfoxide; 2 h, rt Experimental Protocols	Enantioselective Construction of Quinoxaline-Based Heterobiaryls and P,N-Ligands via Chirality Transfer Strategy By: Gao, Zeng; et al Organic Letters (2021), 23(4), 1181-1187.
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Scheme 63 (1 Reaction)

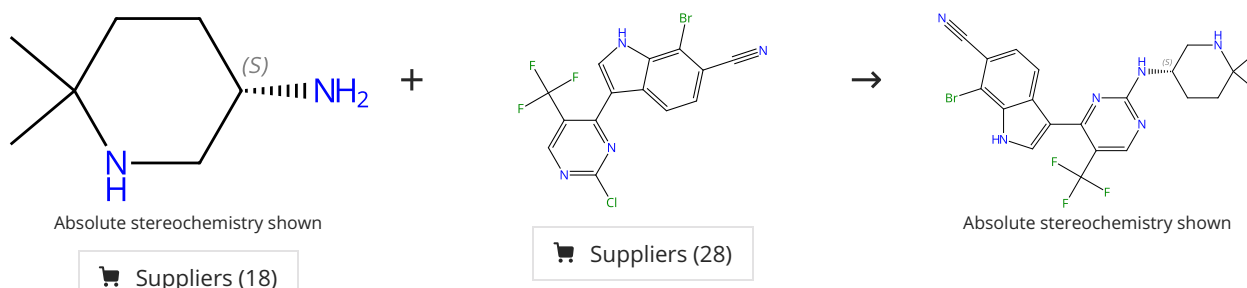
Steps: 1 Yield: 100%



31-113-CAS-22302138 Steps: 1 Yield: 100% 1.1 Solvents: Water; 5 min, rt 1.2 Reagents: Sodium hydroxide; 5 min; 24 h, 130 °C Experimental Protocols	p-Substituted Tris(2-pyridylmethyl)amines as Ligands for Highly Active ATRP Catalysts: Facile Synthesis and Characterization By: Enciso, Alan E.; et al Angewandte Chemie, International Edition (2020), 59(35), 14910-14920.
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Scheme 64 (1 Reaction)

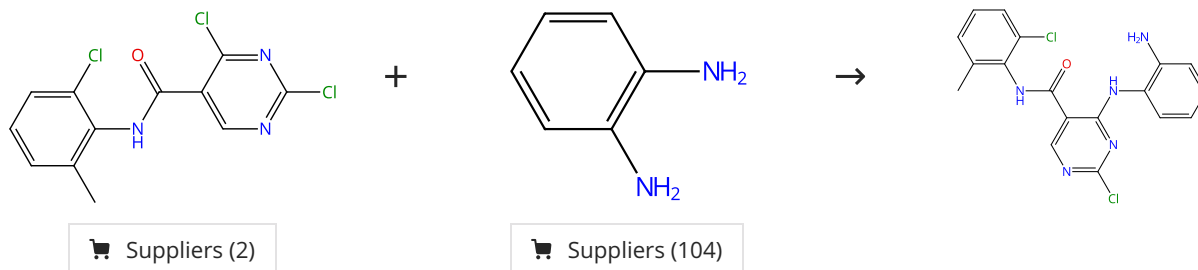
Steps: 1 Yield: 100%



31-614-CAS-40591553	Steps: 1 Yield: 100%	Discovery of SY-5609: A Selective, Noncovalent Inhibitor of CD K7
1.1 Reagents: Diisopropylethylamine Solvents: <i>N</i> -Methyl-2-pyrrolidone; 3 h, 130 °C		By: Marineau, Jason J.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2022), 65(2), 1458-1480.

Scheme 65 (1 Reaction)

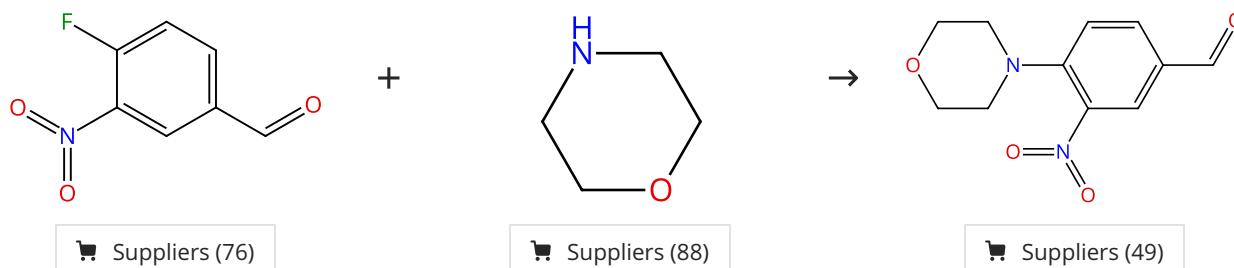
Steps: 1 Yield: 100%



31-614-CAS-23938058	Steps: 1 Yield: 100%	Structure-Based Design of a Potent and Selective Covalent Inhibitor for SRC Kinase That Targets a P-Loop Cysteine
1.1 Reagents: Diisopropylethylamine Solvents: 1-Butanol; 2 h, rt → 110 °C		By: Du, Guangyan; et al
Experimental Protocols		Journal of Medicinal Chemistry (2020), 63(4), 1624-1641.

Scheme 66 (1 Reaction)

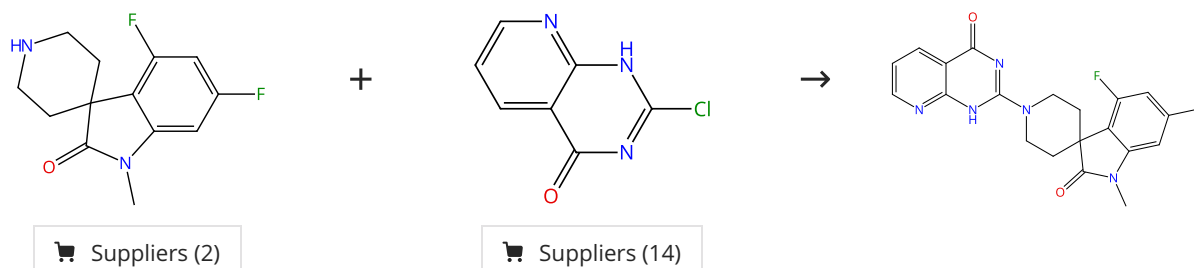
Steps: 1 Yield: 100%



31-614-CAS-32019645	Steps: 1 Yield: 100%	Expansion of the 4-(Diethylamino)benzaldehyde Scaffold to Explore the Impact on Aldehyde Dehydrogenase Activity and Antiproliferative Activity in Prostate Cancer
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 h, 25 °C; 25 °C → rt		By: Ibrahim, Ali I. M.; et al
1.2 Solvents: Water; 30 min, rt		Journal of Medicinal Chemistry (2022), 65(5), 3833-3848.
Experimental Protocols		

Scheme 67 (1 Reaction)

Steps: 1 Yield: 100%



Scheme 68 (1 Reaction) Steps: 1 Yield: 100%

CCOC(=O)C1=C(C(=O)O1)NCCCN2CCOCC2.C1=CC=C(C=C1)Oc2cc(F)c(F)c2O>>CCOC(=O)C1=C(C(=O)O1)NCCCN2CCOCC2.N1=CC=C(C=C1)Oc2cc(F)c(F)c2O1

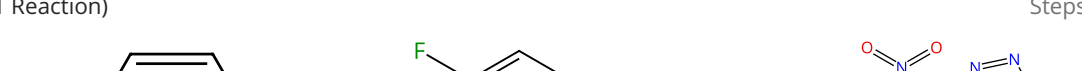
Supplier (1)

Suppliers (75)

Suppliers (2)

Scheme 69 (1 Reaction)

Steps: 1 Yield: 100%


c1ccccc1C2=NN=CN2 + Fc1cc([N+](=O)[O-])cc([N+](=O)[O-])cc1 >> c1ccccc1C2=NN=CN2C#N

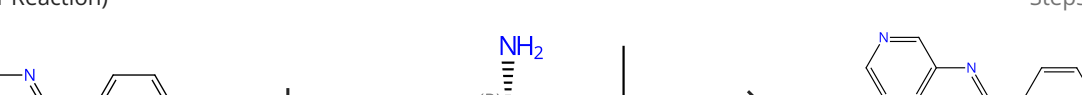
 Suppliers (97)

 Suppliers (75)

 Suppliers (2)

Scheme 70 (1 Reaction)

Steps: 1 Yield: 100%


Chemical reaction scheme showing the synthesis of a complex molecule. The first reactant is a quinoline derivative with a bromine atom and a nitrile group. The second reactant is an amino acid derivative with a chiral center. The product is a complex molecule where the two reactants have reacted, forming a new ring system.

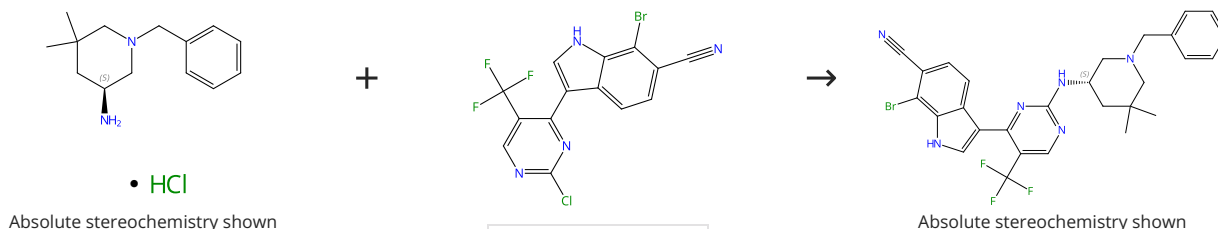
 Suppliers (2)

 Suppliers (52)

31-614-CAS-40188232	Steps: 1 Yield: 100%	Discovery and Characterization of the Potent and Highly Selective 1,7-Naphthyridine-Based Inhibitors BAY-091 and BAY-297 of the Kinase PIP4K2A By: Wortmann, Lars; et al Journal of Medicinal Chemistry (2021), 64(21), 15883-15911.
1.1 Reagents: Triethylamine Solvents: Dimethylformamide; 24 h, rt		

Scheme 71 (1 Reaction)

Steps: 1 Yield: 100%



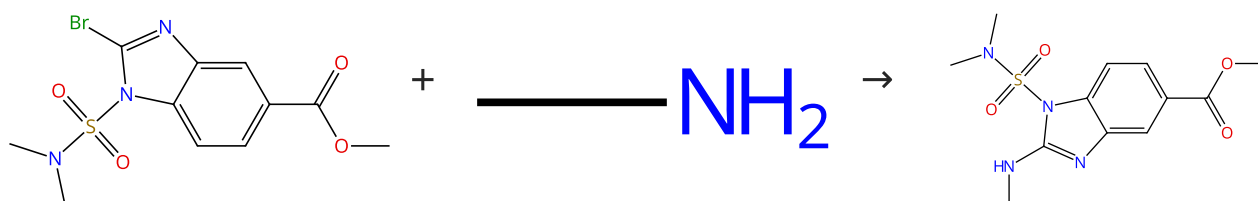
Suppliers (4)

Suppliers (28)

31-614-CAS-40591502	Steps: 1 Yield: 100%	Discovery of SY-5609: A Selective, Noncovalent Inhibitor of CD K7 By: Marineau, Jason J.; et al Journal of Medicinal Chemistry (2022), 65(2), 1458-1480.
1.1 Reagents: Diisopropylethylamine Solvents: <i>N</i> -Methyl-2-pyrrolidone; 4 h, 130 °C Experimental Protocols		

Scheme 72 (1 Reaction)

Steps: 1 Yield: 100%

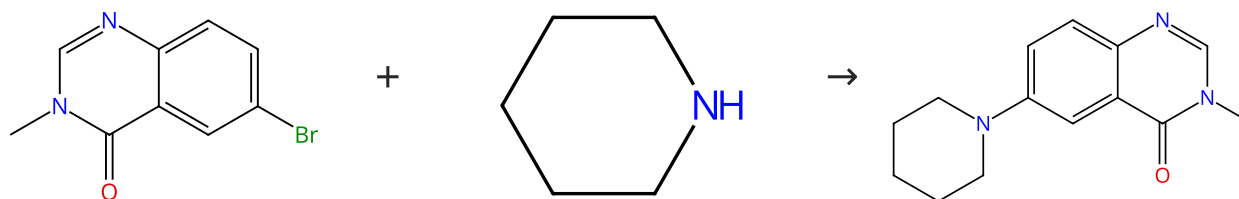


Suppliers (115)

31-172-CAS-22676038	Steps: 1 Yield: 100%	Fragment Screening Hit Draws Attention to a Novel Transient Pocket Adjacent to the Recognition Site of the tRNA-Modifying Enzyme TGT By: Hassaan, Engi; et al Journal of Medicinal Chemistry (2020), 63(13), 6802-6820.
1.1 Solvents: Methanol, Ethyl acetate; 14 h, 23 °C		

Scheme 73 (1 Reaction)

Steps: 1 Yield: 100%



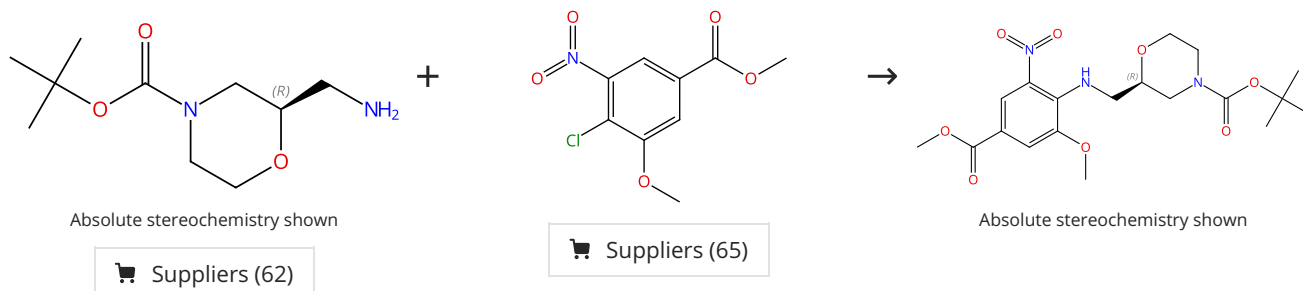
Suppliers (60)

Suppliers (54)

31-614-CAS-36811495 Steps: 1 Yield: 100% 1.1 Reagents: Dabco Catalysts: Nickel bromide (NiBr₂), trihydrate, Osmium(2+), tris (1,10-phenanthroline-κN¹,κN¹⁰)-, (OC-6-11)-, hexafluoro phosphate(1-)(1:2) Solvents: Dimethyl sulfoxide; 18 h, rt Experimental Protocols	Overcoming Photochemical Limitations in Metallaphotoredox Catalysis: Red-Light-Driven C-N Cross-Coupling By: Goldschmid, Samantha L.; et al Journal of the American Chemical Society (2022), 144(49), 22409-22415.
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Scheme 74 (1 Reaction)

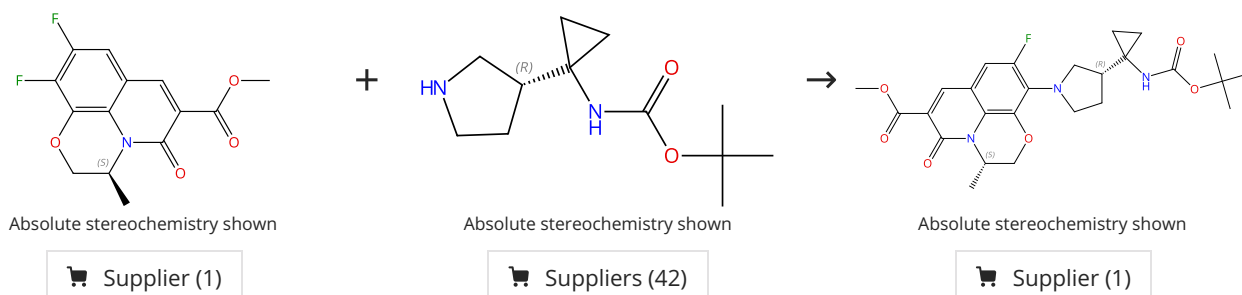
Steps: 1 Yield: 100%



31-614-CAS-35213785 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 19 h, 60 °C Experimental Protocols	Cell-Based Drug Discovery: Identification and Optimization of Small Molecules that Reduce c-MYC Protein Levels in Cells By: Medina, Jesus R.; et al Journal of Medicinal Chemistry (2021), 64(21), 16056-16087.
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Scheme 75 (1 Reaction)

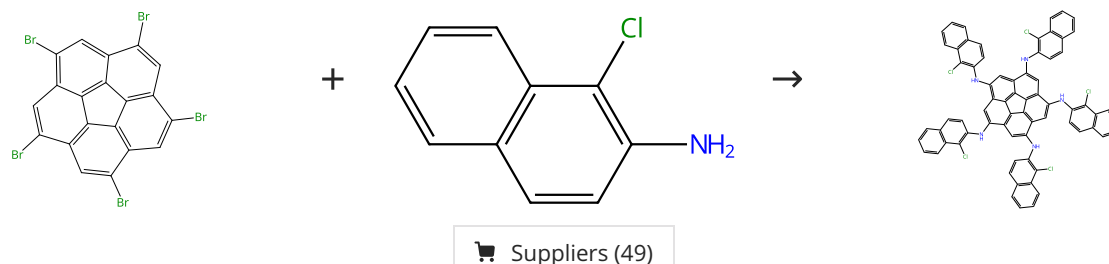
Steps: 1 Yield: 100%



31-614-CAS-40674748 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Dimethyl sulfoxide; 4 h, 90 °C; 90 °C → 40 °C 1.2 Solvents: Water; rt; 16 h, rt Experimental Protocols	Discovery and Optimization of DNA Gyrase and Topoisomerase IV Inhibitors with Potent Activity against Fluoroquinolone-Resistant Gram-Positive Bacteria By: Lapointe, Guillaume; et al Journal of Medicinal Chemistry (2021), 64(9), 6329-6357.
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Scheme 76 (1 Reaction)

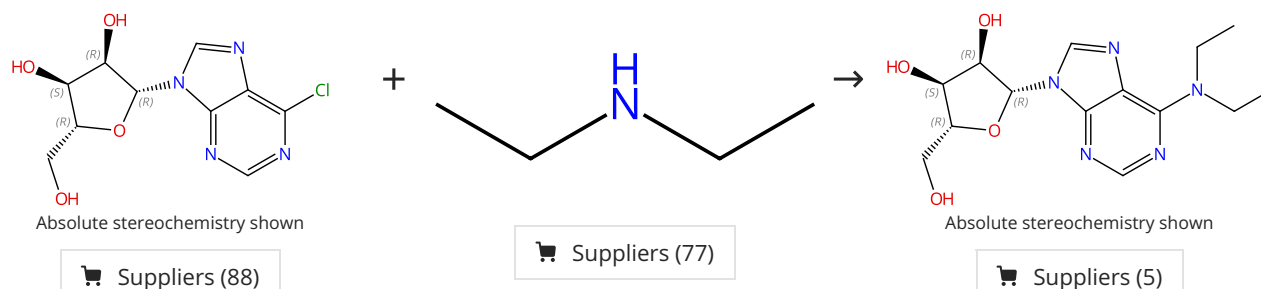
Steps: 1 Yield: 100%



31-614-CAS-31071597 Steps: 1 Yield: 100% 1.1 Reagents: Sodium <i>tert</i>-butoxide Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloro methane adduct Solvents: Toluene; 68 h, 110 °C Experimental Protocols	Five-Fold Symmetric Pentaindolo- and Pentakis(benzoindolo) Corannulenes: Unique Structural Dynamics Derived from the Combination of Helical and Bowl Inversions By: Kise, Koki; et al Angewandte Chemie, International Edition (2022), 61(1), e202112589.
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Scheme 77 (1 Reaction)

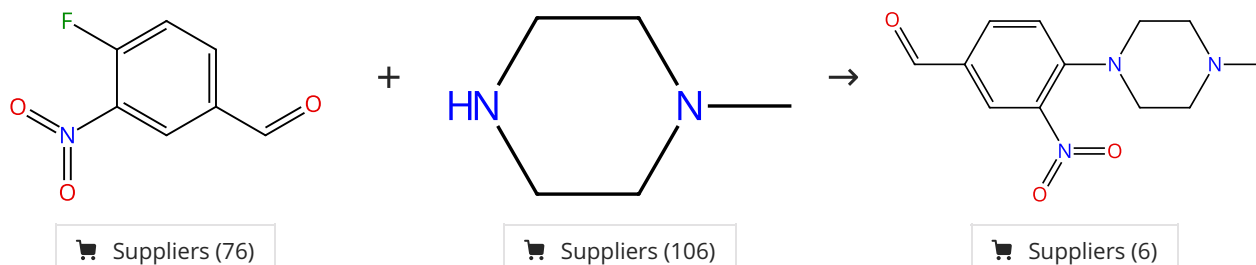
Steps: 1 Yield: 100%



31-614-CAS-34772001 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol, Water; 2 h, rt; 3 h, 60 °C Experimental Protocols	Synthesis of Fluorescent Probes Targeting Tumor-Suppressor Protein FHIT and Identification of Apoptosis-Inducing FHIT Inhibitors By: Kawaguchi, Mitsuyasu; et al Journal of Medicinal Chemistry (2021), 64(13), 9567-9576.
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Scheme 78 (1 Reaction)

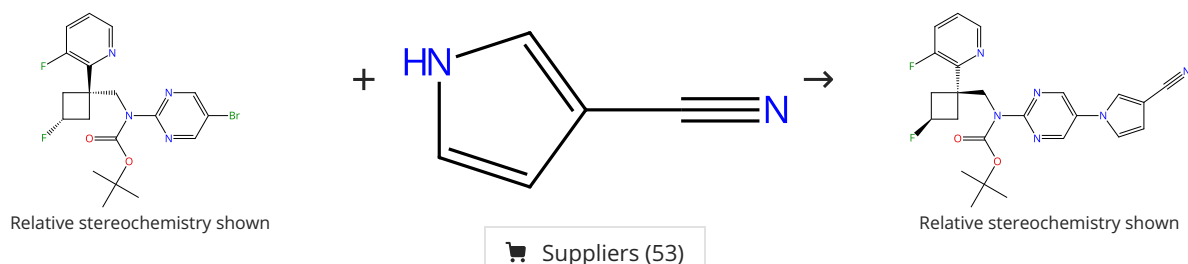
Steps: 1 Yield: 100%



31-614-CAS-32019635 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 1 h, 25 °C; 25 °C → rt 1.2 Solvents: Water; 30 min, rt Experimental Protocols	Expansion of the 4-(Diethylamino)benzaldehyde Scaffold to Explore the Impact on Aldehyde Dehydrogenase Activity and Antiproliferative Activity in Prostate Cancer By: Ibrahim, Ali I. M.; et al Journal of Medicinal Chemistry (2022), 65(5), 3833-3848.
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Scheme 79 (1 Reaction)

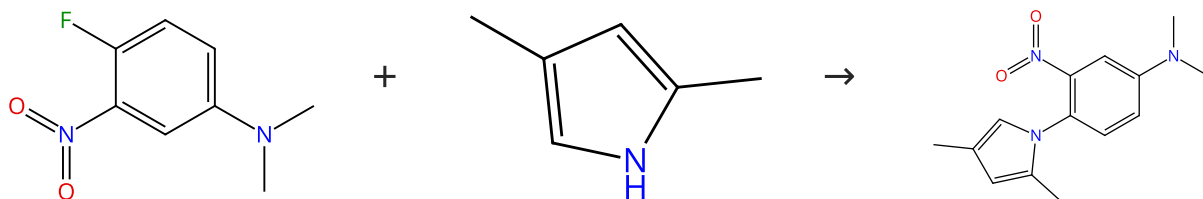
Steps: 1 Yield: 100%



31-614-CAS-31318099 Steps: 1 Yield: 100% 1.1 Reagents: Tripotassium phosphate Catalysts: <i>trans</i>-1,2-Diaminocyclohexane, Cuprous iodide Solvents: Toluene; 2 h, 100 °C Experimental Protocols	Discovery of Reldesemtiv, a Fast Skeletal Muscle Troponin Activator for the Treatment of Impaired Muscle Function By: Collibee, Scott E.; et al Journal of Medicinal Chemistry (2021), 64(20), 14930-14941.
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Scheme 80 (1 Reaction)

Steps: 1 Yield: 100%



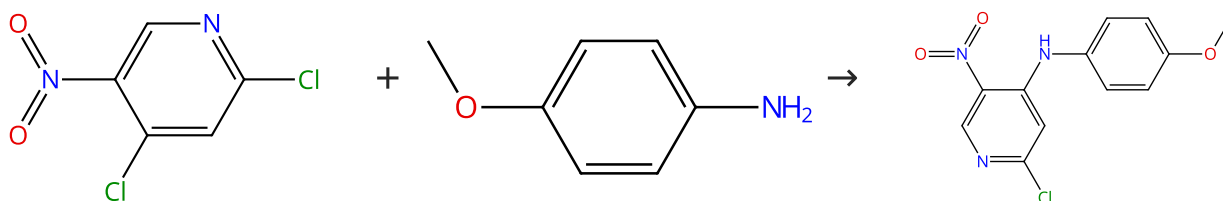
Suppliers (49)

Suppliers (77)

31-172-CAS-23257285 Steps: 1 Yield: 100% 1.1 Reagents: Sodium hydroxide Solvents: Dimethyl sulfoxide; 2 h, rt Experimental Protocols	Enantioselective Construction of Quinoxaline-Based Heterobiaryls and P,N-Ligands via Chirality Transfer Strategy By: Gao, Zeng; et al Organic Letters (2021), 23(4), 1181-1187.
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Scheme 81 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

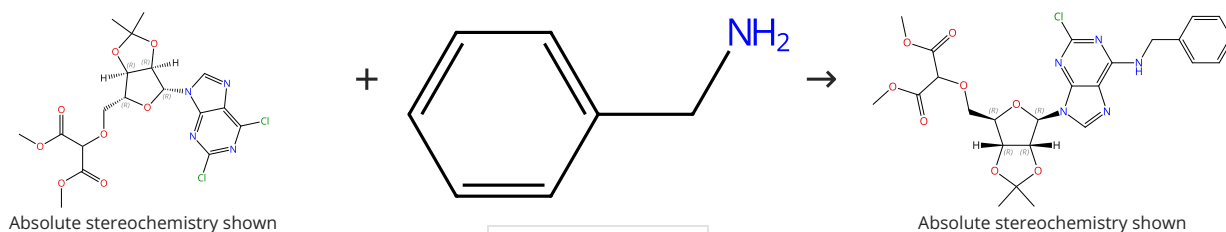
Suppliers (83)

Suppliers (2)

31-614-CAS-31332682 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine Solvents: Acetonitrile; rt Experimental Protocols	A Phenotypic Screen Identifies a Compound Series That Induces Differentiation of Acute Myeloid Leukemia Cells In Vitro and Shows Antitumor Effects In Vivo By: Josa-Cullere, Laia; et al Journal of Medicinal Chemistry (2021), 64(21), 15608-15628.
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Scheme 82 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Supplier (1)

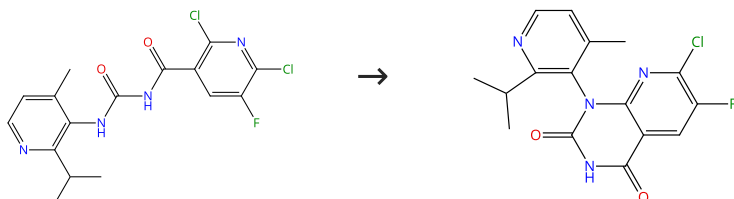
Suppliers (81)

Supplier (1)

31-614-CAS-24298311	Steps: 1 Yield: 100%	Discovery of Potent and Selective Methylenephosphonic Acid CD73 Inhibitors
1.1 Reagents: Triethylamine Solvents: Ethanol; overnight, rt		By: Sharif, Ehesan U.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(1), 845-860.

Scheme 83 (1 Reaction)

Steps: 1 Yield: 100%



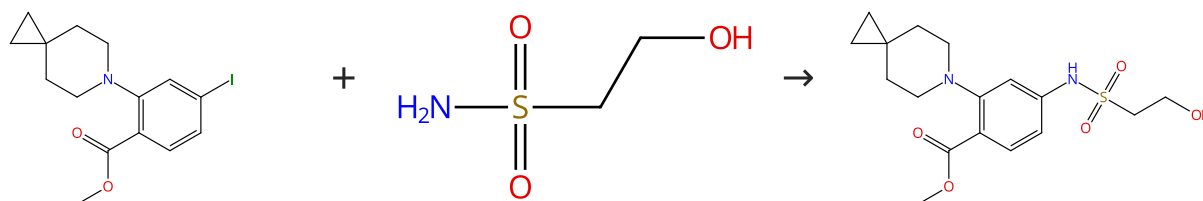
Suppliers (29)

Suppliers (45)

31-172-CAS-21491850	Steps: 1 Yield: 100%	Discovery of a Covalent Inhibitor of K RAS^{G12C} (AMG 510) for the Treatment of Solid Tumors
1.1 Reagents: Potassium bis(trimethylsilyl)amide Solvents: Tetrahydrofuran; 0 °C; 40 min, rt		By: Lanman, Brian A.; et al
1.2 Reagents: Ammonium chloride Solvents: Water; rt		Journal of Medicinal Chemistry (2020), 63(1), 52-65.

Scheme 84 (1 Reaction)

Steps: 1 Yield: 100%



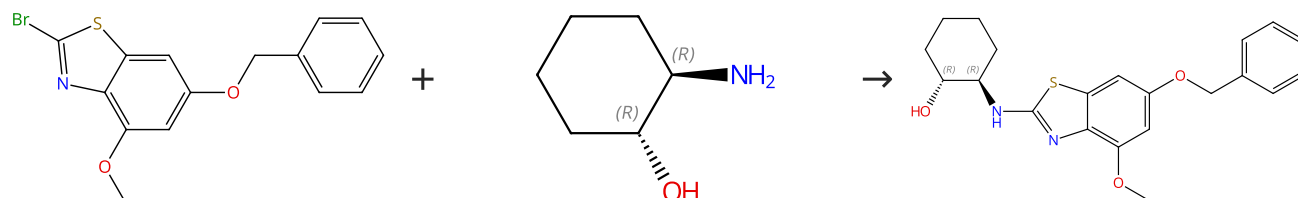
Suppliers (19)

Suppliers (55)

31-614-CAS-34105844	Steps: 1 Yield: 100%	Targeting the Mitotic Kinesin KIF18A in Chromosomally Unstable Cancers: Hit Optimization Toward an In Vivo Chemical Probe
1.1 Reagents: Tripotassium phosphate, <i>N</i> -Hexadecyl- <i>N</i> -methylglycine ethyl ester Catalysts: Cuprous iodide Solvents: Dimethylformamide; 3 h, 100 °C		By: Tamayo, Nuria A.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2022), 65(6), 4972-4990.

Scheme 85 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)

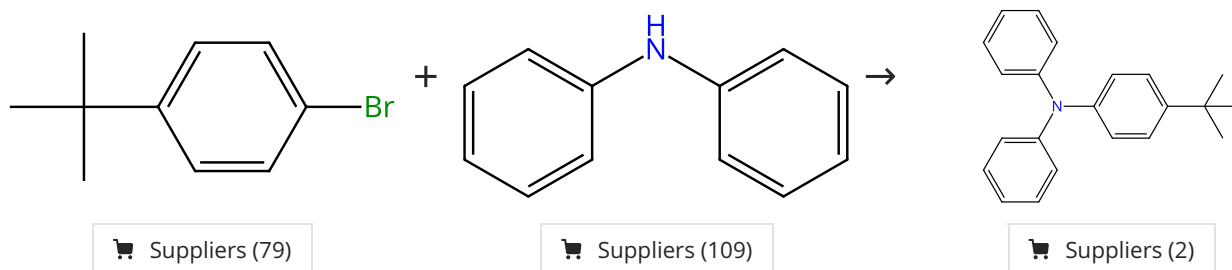
Absolute stereochemistry shown

Suppliers (70)

31-172-CAS-22809321 Steps: 1 Yield: 100%	Discovery of IACS-9439, a Potent, Exquisitely Selective, and Orally Bioavailable Inhibitor of CSF1R
1.1 Reagents: Diisopropylethylamine Solvents: Dimethylformamide; 48 h, 110 °C; 110 °C → rt	By: Czako, Barbara; et al Journal of Medicinal Chemistry (2020), 63(17), 9888-9911.

Scheme 86 (1 Reaction)

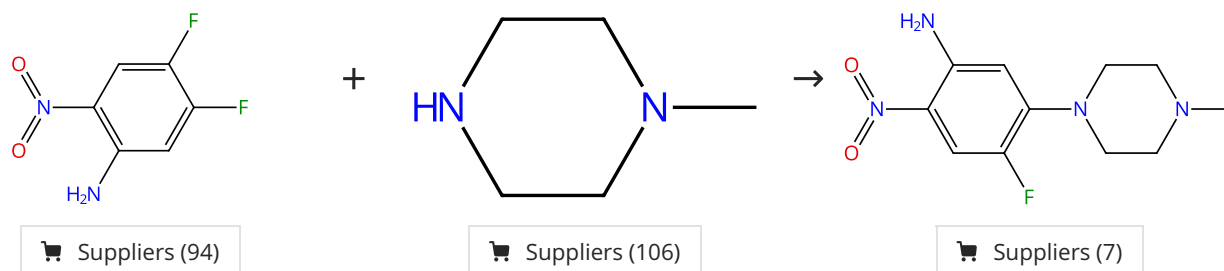
Steps: 1 Yield: 100%



31-172-CAS-22889983 Steps: 1 Yield: 100%	Catalytic and Aerobic Oxidative Biaryl Coupling of Anilines Using a Recyclable Heterogeneous Catalyst for Synthesis of Benzidines and Bicarbazoles
1.1 Reagents: Potassium <i>tert</i> -butoxide Catalysts: Bis(tri- <i>tert</i> -butylphosphine)palladium Solvents: Toluene; rt; 4 h, reflux	By: Matsumoto, Kenji; et al Journal of Organic Chemistry (2020), 85(23), 15154-15166.
Experimental Protocols	

Scheme 87 (1 Reaction)

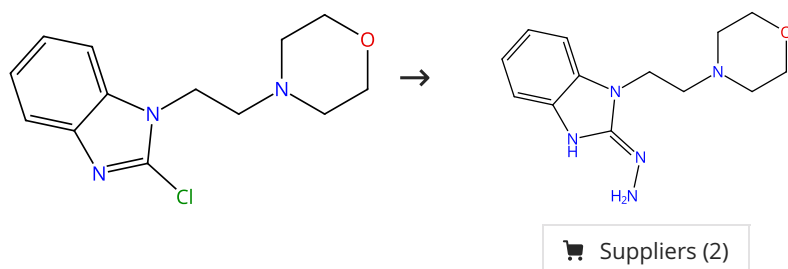
Steps: 1 Yield: 100%



31-172-CAS-22958646 Steps: 1 Yield: 100%	Discovery and optimization of novel pyrazole-benzimidazole CPL304110 as a potent and selective inhibitor of fibroblast growth factor receptors FGFR (1-3)
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 130 °C	By: Yamani, Abdellah; et al European Journal of Medicinal Chemistry (2021), 210, 112990.
Experimental Protocols	

Scheme 88 (1 Reaction)

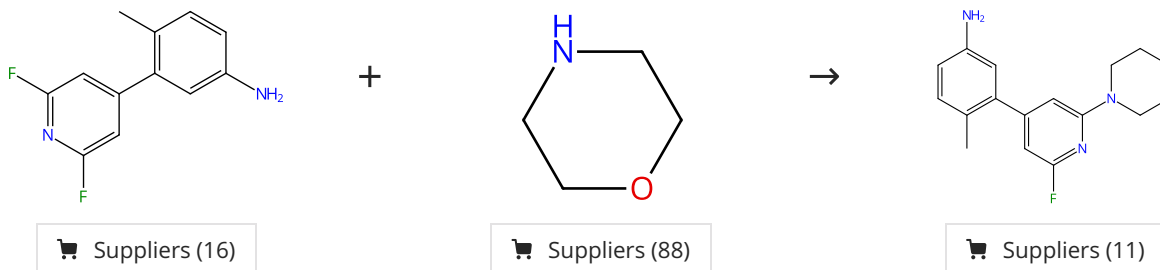
Steps: 1 Yield: 100%



31-614-CAS-31542254 Steps: 1 Yield: 100% 1.1 Reagents: Hydrazine hydrate (1:1) Solvents: Ethylene glycol; 2 h, 140 °C Experimental Protocols	Benzothiazolyl and Benzoxazolyl Hydrazones Function as Zinc Metallochaperones to Reactivate Mutant p53 By: Gilleran, John A.; et al Journal of Medicinal Chemistry (2021), 64(4), 2024-2045.
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Scheme 89 (1 Reaction)

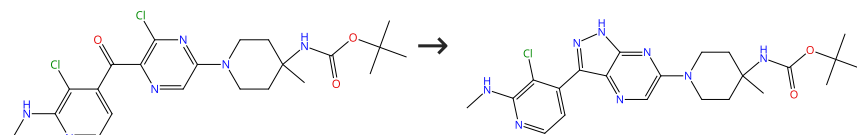
Steps: 1 Yield: 100%



31-172-CAS-21642314 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 3 h, 40 °C	Design and Discovery of N-(3-(2-(2-Hydroxyethoxy)-6-morpholinopyridin-4-yl)-4-methylphenyl)-2-(trifluoromethyl)isonicotinamide, a Selective, Efficacious, and Well-Tolerated RAS AF Inhibitor Targeting RAS Mutant Cancers: The Path to the Clinic By: Ramurthy, Savithri; et al Journal of Medicinal Chemistry (2020), 63(5), 2013-2027.
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Scheme 90 (1 Reaction)

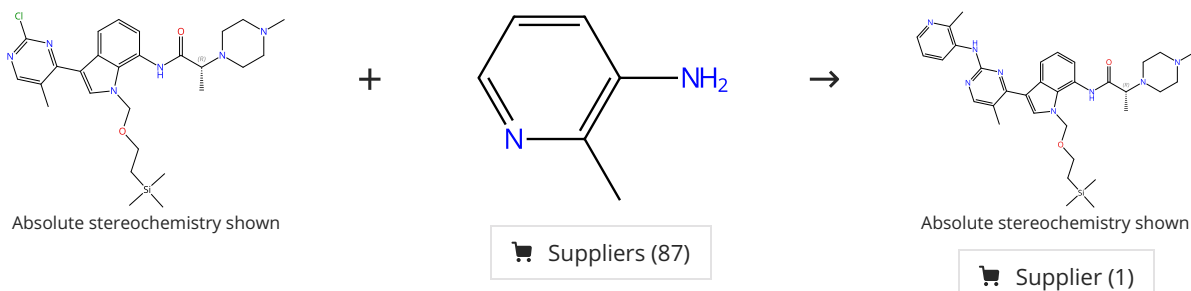
Steps: 1 Yield: 100%



31-614-CAS-32086534 Steps: 1 Yield: 100% 1.1 Reagents: Hydrazine hydrate (1:1) Solvents: Ethanol; 2 h, reflux Experimental Protocols	Discovery of 6-[(3S,4S)-4-Amino-3-methyl-2-oxa-8-azaspiro[4.5]decan-8-yl]-3-(2,3-dichlorophenyl)-2-methyl-3,4-dihydropyrimidin-4-one (IACS-15414), a Potent and Orally Bioavailable SHP2 Inhibitor By: Czako, Barbara; et al Journal of Medicinal Chemistry (2021), 64(20), 15141-15169.
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Scheme 91 (1 Reaction)

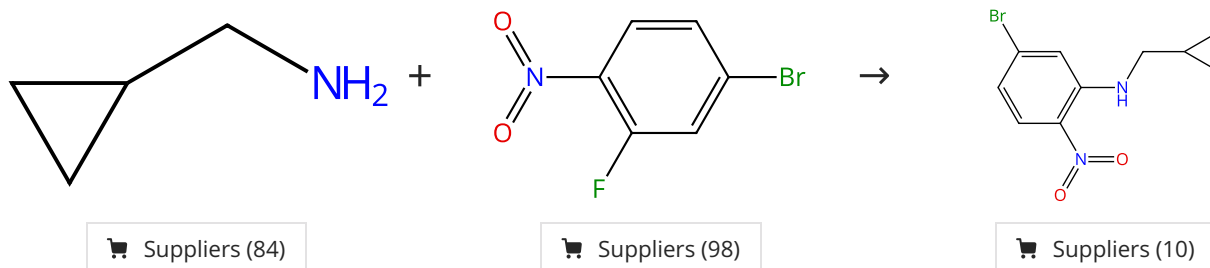
Steps: 1 Yield: 100%



31-172-CAS-22258098	Steps: 1 Yield: 100%	Discovery of (2R)-N-[3-[2-[(3-Methoxy-1-methyl-pyrazol-4-yl)amino]pyrimidin-4-yl]-1H-indol-7-yl]-2-(4-methylpiperazin-1-yl)propenamide (AZD4205) as a Potent and Selective Janus Kinase 1 Inhibitor
1.1 Reagents: Cesium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: 1,4-Dioxane; 2 h, rt → 110 °C		By: Su, Qibin; et al Journal of Medicinal Chemistry (2020), 63(9), 4517-4527.
Experimental Protocols		

Scheme 92 (1 Reaction)

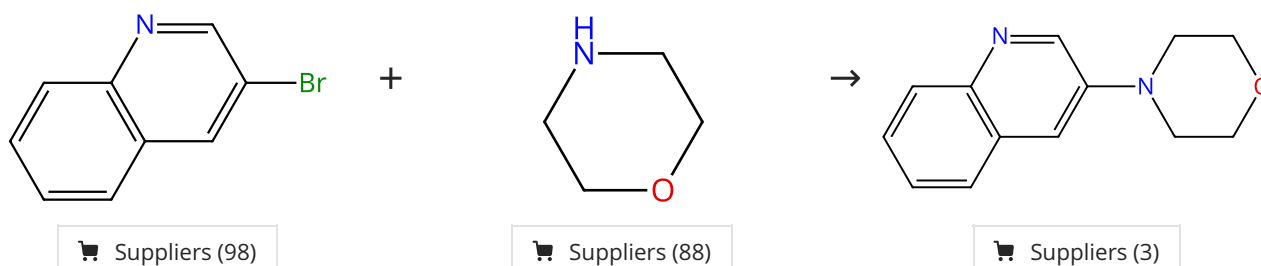
Steps: 1 Yield: 100%



31-614-CAS-34289453	Steps: 1 Yield: 100%	Rational Design of Original Fused-Cycle Selective Inhibitors of Tryptophan 2,3-Dioxygenase
1.1 Reagents: Triethylamine Solvents: Ethanol; 30 - 960 min, reflux		By: Kozlova, Arina; et al Journal of Medicinal Chemistry (2021), 64(15), 10967-10980.
Experimental Protocols		

Scheme 93 (1 Reaction)

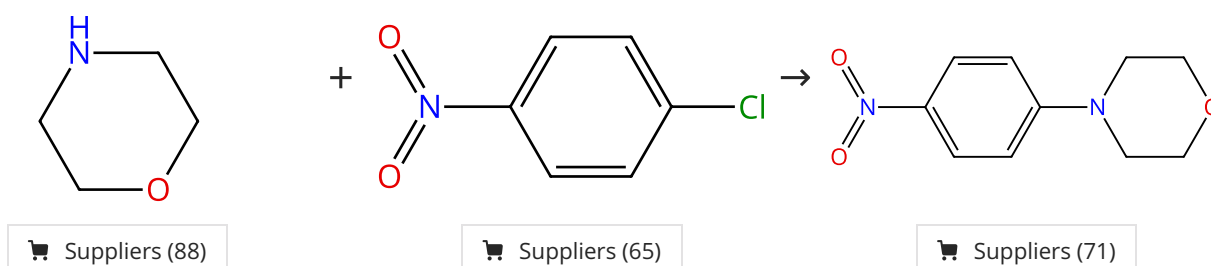
Steps: 1 Yield: 100%



31-614-CAS-25850251	Steps: 1 Yield: 100%	A Neophyl Palladacycle as an Air- and Thermally Stable Precursor to Oxidative Addition Complexes
1.1 Reagents: Sodium <i>tert</i> -butoxide Catalysts: [(1,2,5,6-η)-1,5-Cyclooctadiene][(2,2-dimethyl-1,2-ethanediyl)-1,2-phenylene]palladium, [2',6'-Bis(1-methyl ethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: Tetrahydrofuran; 16 h, 80 °C		By: King, Ryan P.; et al Organic Letters (2021), 23(20), 7927-7932.
Experimental Protocols		

Scheme 94 (1 Reaction)

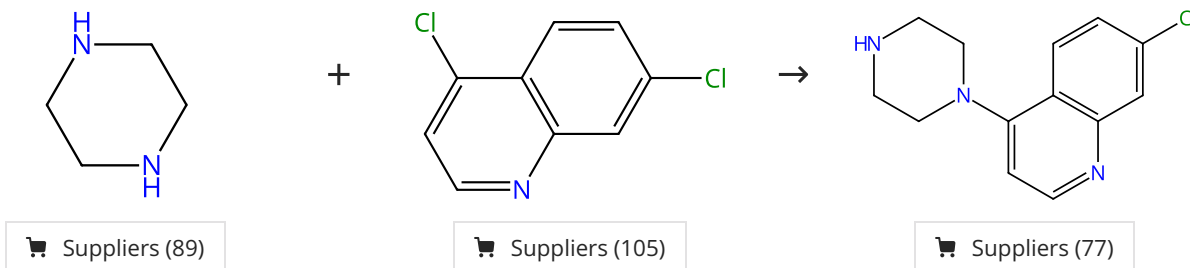
Steps: 1 Yield: 100%



31-172-CAS-21568173	Steps: 1 Yield: 100%	Development, synthesis and biological investigation of a novel class of potent PC-PLC inhibitors
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; 19 h, 120 °C		By: Pilkington, Lisa I.; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2020), 191, 112162pp..

Scheme 95 (1 Reaction)

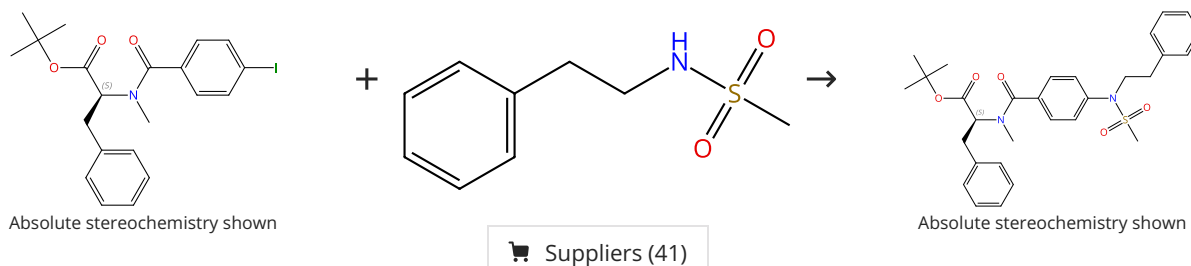
Steps: 1 Yield: 100%



31-172-CAS-23618973	Steps: 1 Yield: 100%	Design and synthesis of quinoline-pyrimidine inspired hybrids as potential plasmodial inhibitors
1.1 Solvents: Isopropanol; 20 h, reflux		By: Kayamba, Francis; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2021), 217, 113330.

Scheme 96 (1 Reaction)

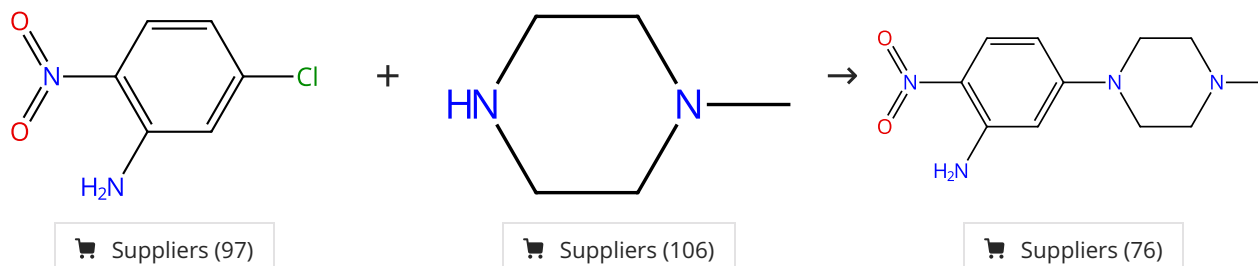
Steps: 1 Yield: 100%



31-614-CAS-24353263	Steps: 1 Yield: 100%	Modulators of hERAP2 discovered by high-throughput screening
1.1 Reagents: Tripotassium phosphate Catalysts: <i>trans</i> -1,2-Diaminocyclohexane, Cuprous iodide Solvents: 1,4-Dioxane; 36 h, 110 °C		By: Laura, Medve; et al
1.2 Reagents: Sodium hydroxide, Ammonia Solvents: Water		European Journal of Medicinal Chemistry (2021), 211, 113053.
Experimental Protocols		

Scheme 97 (1 Reaction)

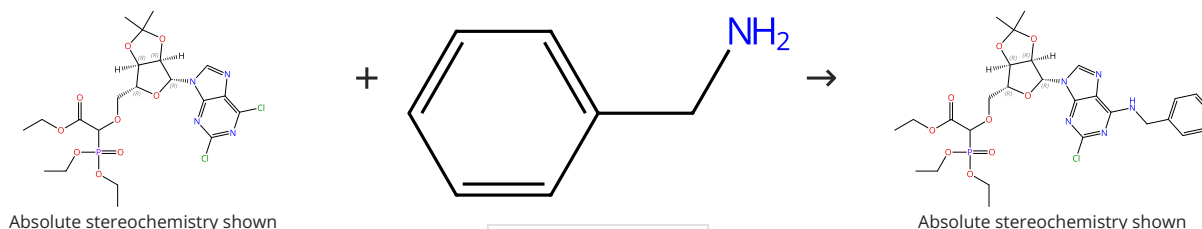
Steps: 1 Yield: 100%



31-172-CAS-22955811	Steps: 1 Yield: 100%	Discovery and optimization of novel pyrazole-benzimidazole CPL304110 as a potent and selective inhibitor of fibroblast growth factor receptors FGFR (1-3)
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; overnight, 130 °C		By: Yamani, Abdellah; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2021), 210, 112990.

Scheme 98 (1 Reaction)

Steps: 1 Yield: 100%

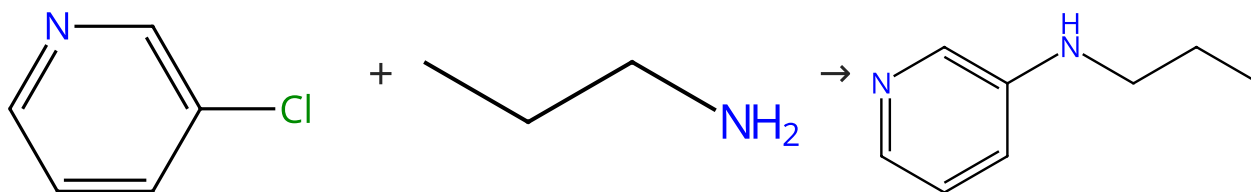


Suppliers (81)

31-614-CAS-24298362	Steps: 1 Yield: 100%	Discovery of Potent and Selective Methylenephosphonic Acid CD73 Inhibitors
1.1 Reagents: Triethylamine Solvents: Ethanol; overnight, rt		By: Sharif, Ehesan U.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2021), 64(1), 845-860.

Scheme 99 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (78)

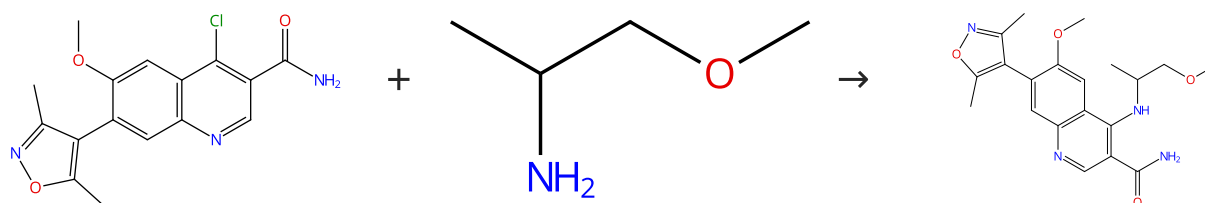
Suppliers (70)

Suppliers (31)

31-172-CAS-22693832	Steps: 1 Yield: 100%	Development of an Aryl Amination Catalyst with Broad Scope Guided by Consideration of Catalyst Stability
1.1 Reagents: Sodium <i>tert</i> -butoxide Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC ^{1'}]phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (<i>SP</i> -4-2) Solvents: Tetrahydrofuran, Dodecane; 1 h, rt		By: McCann, Scott D.; et al
1.2 Reagents: Oxygen Solvents: Ethyl acetate; rt		Journal of the American Chemical Society (2020), 142(35), 15027-15037.

Scheme 100 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (30)

Suppliers (64)

31-614-CAS-35385585	Steps: 1 Yield: 100%	Discovery of a Novel Bromodomain and Extra Terminal Domain (BET) Protein Inhibitor, I-BET282E, Suitable for Clinical Progression By: Jones, Katherine L.; et al Journal of Medicinal Chemistry (2021), 64(16), 12200-12227.
1.1 Reagents: Diisopropylethylamine Solvents: <i>N</i> -Methyl-2-pyrrolidone; 8 h, 120 °C		
Experimental Protocols		

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